

DEPROPANIZER COLUMN VER 1

Operator Training Simulator · C-201

Chulalongkorn University · Department of Chemical Engineering

HAZOP REPORT

Hazard and Operability Study of the C-201 depropanizer

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Teaching document generated from the HAZOP tab of the C-201 simulator. It is not a study of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
A	03-Oct-26	Issued for teaching	A. Soottitantawat			

HAZOP CLOSE-OUT SUMMARY

Project: Depropanizer Column Ver 1 · C-201 Operator Training Simulator · Document DC3-V1-SA-HZP-001 rev A · 36 actions, all open at issue. The action party fills "Action taken" and the HAZOP leader changes the status to CLOSED when the close-out sheet (Appendix 3) is signed.

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1 Feed surge drum V-201 and feed pumps P-201A/B	R01	N1.1, N1.2, N2.1	Put V-201 level on the DCS (rev B LT-201 with LAL/LAH-201) and add a low-low level trip that stops P-201A/B before the suction vortexes (about 10 % in the model); consider the rev C cascade LIC-201 → FIC-201 so the drum level sets the feed.	Instrument / Process		OPEN	Risk 12 High
N1/N5/N6 Feed surge drum V-201 and feed pumps P-201A/B / Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201 / Reflux drum V-202, pumps P-202A/B and propane product LV-203	R02	N1.2, N1.6, N5.8, N6.7	Relief and fire review: confirm the design pressures of V-201, V-202 and the E-201 process side (not on the P&ID); give V-201 its own PSV sized for fire, and confirm V-202 is protected by PSV-201 with no isolation between; add fireproofing, water deluge and a depressuring route for the LPG vessels (BLEVE prevention).	Process / Safety		OPEN	Risk 12 High
N1 Feed surge drum V-201 and feed pumps P-201A/B	R03	N1.1, N1.3, N2.3	Bring FSLL-201 to the console as an alarm (today it only starts P-201B) and agree with the upstream unit how a feed stop is announced in both directions.	Instrument / Operations		OPEN	Risk 12 High
N1/N6/N7 Feed surge drum V-201 and feed pumps P-201A/B / Reflux drum V-202, pumps P-202A/B and propane product LV-203 / Bottoms to the debutanizer (LIC-202, LV-202)	R04	N1.4, N6.5, N7.8	Put the check valves on the P-201A/B and P-202A/B discharges into the inspection plan (a passing check valve recirculates through the idle pump); add a check valve on the bottoms line downstream of LV-202 (recommended addition).	Mechanical		OPEN	Risk 6 Medium
N1/N6 Feed surge drum V-201 and feed pumps P-201A/B / Reflux drum V-202, pumps P-202A/B and propane product LV-203	R05	N1.5, N2.3, N6.4	Provide minimum-flow recirculation lines with restriction orifices from the P-201A/B and P-202A/B discharges back to V-201 and V-202, so a closed FV-201/XV-201, or FV-204 with LV-203, cannot deadhead the running pump.	Process		OPEN	Risk 9 Medium
N1/N9 Feed surge drum V-201 and feed pumps P-201A/B / Procedures: start-up, shutdown, emergency shutdown, maintenance	R06	N1.7, N9.5	Check the minimum design metal temperature of V-201, V-202, C-201 and the LPG lines against auto-refrigeration: released or depressured LPG flashes toward its atmospheric boiling point (about –42 °C for propane). Depressure slowly, never re-pressure a cold vessel, add a low-temperature alarm on TI-201.	Mechanical / Process		OPEN	Risk 8 Medium
N1/N6/N9 Feed surge drum V-201 and feed pumps P-201A/B / Reflux drum V-202, pumps P-202A/B and propane product LV-203 / Procedures: start-up, shutdown, emergency shutdown, maintenance	R07	N1.8, N6.8, N9.6	Water draining of the LPG drums: water boots with interface level on V-201 and V-202 (recommended addition), double-block drains with a spring-return valve, drains always attended; hydrate and ice awareness at –1 °C.	Process / Operations		OPEN	Risk 10 High

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
all all	R08	N1.9, N3.9, N5.8, N6.2, N6.9, N9.7	Gas detection around P-201A/B, P-202A/B, V-201, V-202, E-203 and the bottoms line, with remote-operated shut-off valves on the V-201 and V-202 outlets and on the column bottoms; double mechanical seals with seal-leak alarms on the LPG pumps.	Instrument / Mechanical		OPEN	Risk 10 High
N2 Feed control FV-201 and preheater E-201 to tray 18	R09	N2.6	Thermal relief on the E-201 process side (recommended addition), or a locked procedure: steam off (XV-202, TV-202) before the feed is blocked in.	Process		OPEN	Risk 8 Medium
N2/N3 Feed control FV-201 and preheater E-201 to tray 18 / C-201 stripping section (trays 1-17), sump and reboiler E-203	R10	N2.7, N3.10	Hydrocarbon detection on the E-201 and E-203 steam condensate: the process side is at higher pressure than the steam side, so a tube leak puts LPG into the condensate return. Route the condensate through a flash pot vented to a safe place.	Process / Instrument		OPEN	Risk 8 Medium
N3/N4/N5 C-201 stripping section (trays 1-17), sump and reboiler E-203 / C-201 rectifying section (trays 18-27) and reflux FV-204 / Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	R11	N3.1, N4.2, N5.1, N5.2, N5.3, N5.6, N9.4	High-pressure trip PSHH on two or three independent transmitters (recommended addition, 2oo3) that closes XV-209 and XV-202 (heat off) before PSV-201 lifts; LOPA to set its SIL. Today PSV-201 is the only layer against every overpressure scenario.	Process safety / Instrument		OPEN	Risk 15 High
N3 C-201 stripping section (trays 1-17), sump and reboiler E-203	R12	N2.5, N3.2, N7.5, N8.3	Reboiler margin: alarm on TV-208 opening above 90 % and trend PI-205 (steam chest) so fouling, low header pressure and a stuck valve are seen before the bottoms go off spec.	Instrument / Operations		OPEN	Risk 8 Medium
N3/N7 C-201 stripping section (trays 1-17), sump and reboiler E-203 / Bottoms to the debutanizer (LIC-202, LV-202)	R13	N3.4, N7.2, N7.7	On low sump level close the bottoms line too: LSHH-202 today closes only the steam valves, so vapour at 1,718 kPa g can blow through LV-202 into the debutanizer at 480 kPa g. Add an XV on the bottoms line (recommended addition) and check the debutanizer relief for gas blow-by.	Process / Instrument		OPEN	Risk 8 Medium
N3/N7 C-201 stripping section (trays 1-17), sump and reboiler E-203 / Bottoms to the debutanizer (LIC-202, LV-202)	R14	N3.5, N7.1, N7.3, N7.7	Independent high-high sump level switch LSHH-202 (recommended addition) with alarm, and a procedure to cut feed and heat when the sump cannot be emptied.	Instrument		OPEN	Risk 9 Medium
N3 C-201 stripping section (trays 1-17), sump and reboiler E-203	R15	N3.1, N3.8	Deviation alarm between TIC-208 (tray 8) and the neighbouring TI-242 (tray 4) and TI-243 (tray 12) so a drifting control transmitter is seen.	Instrument		OPEN	Risk 9 Medium
N1/N4/N6 Feed surge drum V-201 and feed pumps P-201A/B / C-201 rectifying section (trays 18-27) and reflux FV-204 / Reflux drum V-202, pumps P-202A/B and propane product LV-203	R16	N1.3, N4.1, N6.3	Test the auto-start of P-201B (FSLL-201) and P-202B (PDSL-230) at a fixed interval and after maintenance; record the tests.	Operations / Instrument		OPEN	Risk 9 Medium
N4 C-201 rectifying section (trays 18-27) and reflux FV-204	R17	N4.2, N4.4	Reflux too high: FIC-204 HI alarm response note and a high limit on the FIC-204 set point; review the fail-open choice of FV-204 (the model floods the column and lifts PSV-201 when it fails open).	Process / Instrument		OPEN	Risk 12 High

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N4/N6/N7 C-201 rectifying section (trays 18-27) and reflux FV-204 / Reflux drum V-202, pumps P-202A/B and propane product LV-203 / Bottoms to the debutanizer (LIC-202, LV-202)	R18	N2.1, N2.8, N3.6, N3.7, N4.3, N4.6, N6.6, N7.5	Validate AI-219 and AI-220 against a laboratory sample every shift; alarm on analyzer fault.	Laboratory / Instrument		OPEN	Risk 8 Medium
N4 C-201 rectifying section (trays 18-27) and reflux FV-204	R19	N4.7	Composition control: run the TIC-227 → FIC-204 cascade and TIC-208 together only after a decoupling review; operating note on which loop to put in MAN first.	Process control		OPEN	Risk 4 Low
N5/N8 Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201 / Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	R20	N5.1, N5.3, N6.7, N8.2, N8.7	Flare study: loss of cooling water is common to every column on the site, so check the flare header and flare for the simultaneous relief (C-201 alone reaches about 19,900 Nm ³ /h in the model); back-pressure on PSV-201.	Process safety		OPEN	Risk 15 High
N5/N6 Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201 / Reflux drum V-202, pumps P-202A/B and propane product LV-203	R21	N5.1, N6.2	Low-low level on V-202 (LSLL-203, recommended addition) that stops P-202A/B before they run dry.	Instrument		OPEN	Risk 15 High
N5 Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	R22	N5.2	Car-seal HV-210 open (CSO register) and add a PIC-210 output-high alarm: a vent that cannot open leaves the column to PSV-201.	Operations / Instrument		OPEN	Risk 12 High
N5/N8 Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201 / Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	R23	N5.4, N5.5, N6.1, N8.6	Fuel-gas interface: knock-out drum on the PV-210 vent before the fuel-gas header, high-flow alarm on FI-217, and tell the fuel-gas system owner that PV-210 fails open.	Process		OPEN	Risk 9 Medium
N5 Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	R24	N5.6	Deviation alarm between PIC-210 and PI-211: two independent transmitters must agree; operating note on which to believe.	Instrument		OPEN	Risk 8 Medium
N5 Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	R25	N5.7	PSV-201 inlet and outlet isolation valves car-sealed open; test and inspection interval; flare line free-draining.	Mechanical / Operations		OPEN	Risk 5 Medium
N5 Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	R26	N5.9, N6.8	Hydrocarbon detection on the E-202 cooling-water return: LPG leaking into the cooling water is released at the cooling tower.	Instrument		OPEN	Risk 8 Medium
N6 Reflux drum V-202, pumps P-202A/B and propane product LV-203	R27	N6.1	Let LSHH-203 also close PV-210 (or confirm the knock-out drum of R23) so liquid LPG cannot reach the fuel-gas header when V-202 overfills.	Instrument / Process		OPEN	Risk 8 Medium

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N6 Reflux drum V-202, pumps P-202A/B and propane product LV-203	R28	N4.6, N6.6	Off-spec propane: divert the product to an off-spec tank on AI-219 HI instead of the sales tank.	Process / Operations		OPEN	Risk 8 Medium
N8 Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	R29	N5.5, N8.1	Instrument-air failure: review the fail positions (PV-210 fails open to fuel gas, LV-202 and LV-203 fail closed) and consider an ESD on low air-header pressure; confirm the air receiver hold-up time.	Instrument / Process		OPEN	Risk 9 Medium
N8 Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	R30	N8.5	Power failure: supply P-202B from a different bus (or a steam driver like P-201B) and write the power-failure response procedure.	Electrical / Operations		OPEN	Risk 9 Medium
N8 Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	R31	N8.4	Confirm the steam-side design pressure of E-201 and E-203 against the highest LP header pressure, and their relief.	Mechanical		OPEN	Risk 6 Medium
N9 Procedures: start-up, shutdown, emergency shutdown, maintenance	R32	N9.3	Start-up: purge C-201, V-202, V-201 and the lines with nitrogen to an oxygen content below the limit before hydrocarbon is admitted; record the gas tests; hold point in the procedure.	Operations		OPEN	Risk 10 High
N9 Procedures: start-up, shutdown, emergency shutdown, maintenance	R33	N9.8	Turnaround entry: positive isolation (blinds), nitrogen disconnected, oxygen and hydrocarbon tests, standby person and rescue plan for every vessel entry.	Operations / Safety		OPEN	Risk 8 Medium
N9 Procedures: start-up, shutdown, emergency shutdown, maintenance	R34	N9.9	Trip-bypass management for LSLL-202, LSHH-203 and the auto-starts: permit, time limit, bypass shown on the DCS, proof tests at the interval set by LOPA.	Operations / Instrument		OPEN	Risk 8 Medium
N9 Procedures: start-up, shutdown, emergency shutdown, maintenance	R35	N9.10	Steam-out: confirm the vacuum rating of C-201 and V-202, or keep a vent open while the steamed vessel cools.	Mechanical / Operations		OPEN	Risk 8 Medium
all all	R36	N1.6, N1.9, N3.9, N4.1, N4.2, N5.1, N5.2, N5.7, N6.1, N6.9, N7.2, N8.1, N9.3, N9.6	LOPA for the High rows that rely on PSV-201, LSHH-203, LSLL-202 and on the operator (overpressure from cooling-water loss, a blocked vent or too much reflux; V-201 overfill and fire; LPG leaks).	Process safety		OPEN	Risk 15 High

HAZOP REPORT

Hazard and Operability Study of the C-201 depropanizer column
Depropanizer Column Ver 1 · C-201 Operator Training Simulator

	Role	Name	Signature	Date
Prepared by	HAZOP scribe			
Reviewed by	HAZOP leader (chair)			
Approved by	Plant / project manager			

This report records a teaching HAZOP of the column drawn in the simulator. Names, signatures and dates are filled in by the class team that runs the study; the risk matrix is an example for teaching, not a corporate matrix.

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Executive summary

A HAZOP of the C-201 depropanizer, from the feed surge drum V-201 and feed pumps P-201A/B through the preheater E-201, the column with its reboiler E-203, the overhead condenser E-202, vent to fuel gas and PSV-201, the reflux drum V-202 with pumps P-202A/B, the propane and bottoms products, the utilities, the SIS and the procedures, was carried out with the IEC 61882 guide-word method. The unit was divided into **9 nodes**; **78 deviations** were recorded and **36 recommendations** raised.

Nodes	Deviations	Recommendations	Very high	High	Medium	Low
9	78	36	0	13	54	11

Risk is ranked with the existing safeguards credited (Section 3.4). High-ranked deviations are carried to the LOPA (LOPA and SRS of SIF tabs).

Key findings (dynamic check on the simulator)

- **PSV-201 is the only layer against overpressure.** In 3 of the runs (N4 flow more, N5 pressure more, N5 pressure more) the column reaches the relief set point with no trip at all: FV-204 failing open lifts it at 03:02; loss of cooling water lifts it at 00:27; a closed vent lifts it at 07:41. A high-pressure trip that cuts the reboiler steam (R11) would stop the vapour at its source; LOPA must decide its SIL (R36).
- **Cooling-water loss is the governing relief case:** FI-216 LO at 00:02 comes only 25 s before PSV-201 lifts at 00:27; the peak is 1.868 MPa g and up to 19,853 Nm³/h go to flare (6,479 Nm³ in 30 min). Because the whole site loses cooling water together, the flare must take every column at once (R20). Meanwhile V-202 empties to 0.3 % and the reflux pumps run dry (R21).
- **Too much reflux also ends at the PSV:** FV-204 fails open, the column floods (PDI-212 to 112 kPa) and TIC-208 adds steam to hold tray 8, so the extra reflux is boiled back up and the pressure rises. A controller doing its job can drive an upset toward relief (R11, R17).
- **LSHH-203 catches both overfill cases:** with no reflux it trips at 05:07, with the product valve shut at 13:42 (first alarm LIC-203 HI at 09:00). The trip stops feed and heat but leaves PV-210 open, so if it failed nothing else would keep liquid out of the fuel gas; R23 and R27 add that barrier.
- **A full sump gives no trip:** with LV-202 shut the sump reaches 100 % and floods the bottom trays (PDI-212 HI at 19:57); the only protection is the operator on LIC-202 HI (R14).
- **V-201 has no DCS level in rev A:** whenever the feed stops while the upstream unit keeps sending (loss of both pumps, instrument-air failure) the drum passes 95 % after about 11:59 and nothing alarms (R01).
- **Instrument-air failure is safe for pressure but not for inventory:** PV-210 fails open and the column falls to 0.84 MPa g into the fuel gas, the sump fills and V-202 empties, yet no trip acts. Many alarms at once point to a common utility (R29).
- **Slow utility problems show first as quality:** with the LP header falling, PI-207 LO at 00:45 comes first and AI-220 reaches 23.3 %; no safety limit is approached, the loss is product value.

Actions by action party

Action party	Number	Action No.
Instrument	19	R01, R03, R08, R10, R11, R12, R13, R14, R15, R16, R17, R18, R21, R22, R24, R26, R27, R29, R34
Process	13	R01, R02, R05, R06, R07, R09, R10, R13, R17, R23, R27, R28, R29
Operations	12	R03, R07, R12, R16, R22, R25, R28, R30, R32, R33, R34, R35
Mechanical	6	R04, R06, R08, R25, R31, R35
Process safety	3	R11, R20, R36
Safety	2	R02, R33
Laboratory	1	R18
Process control	1	R19
Electrical	1	R30

An action shared by two parties (e.g. Instrument / Operations) is counted under both.

1 Introduction

1.1 Purpose

The HAZOP identifies how C-201 can deviate from its design intent, what the deviations lead to, whether the existing safeguards are adequate, and what should be changed. It also finds operability problems: deviations that do not harm people but put the propane or the bottoms off specification, flare product or waste steam. The study feeds the safety layers, LOPA and SRS of the SIFs, and the cause and effect diagram DC3-V1-IN-CE-001.

1.2 The unit

C-201 is a 27-tray depropanizer. Mixed C3/C4 liquid from the feed surge drum V-201 is pumped by P-201A (P-201B turbine spare) under FIC-201 (FV-201, 60 m³/h) through the LP-steam preheater E-201 (TIC-202, 75 °C) to tray 18. The reboiler E-203 on LP steam (TIC-208 on tray 8, TV-208) boils up from the sump (LIC-202, LV-202 bottoms to the debutanizer at 0.48 MPa g). The overhead is condensed in E-202 on cooling water (330 m³/h); the column top is held at 1.70 MPa g by PIC-210 venting non-condensables to fuel gas, and PSV-201 relieves to flare at 1.80 MPa g. Reflux drum V-202 feeds P-202A/B (P-202B spare): reflux FIC-204 (49 m³/h) to tray 27 and propane product LIC-203 (LV-203, 32 m³/h). The SIS closes the steam valves XV-209 and XV-202 on low-low sump level (LSLL-202), and also the feed valve XV-201 on high-high reflux drum level (LSHH-203) or the ESD push-button; FSLL-201 and PDSL-230 auto-start the spare pumps.

1.3 Objectives

- Identify hazards to people, the environment and the plant from deviations of each node.
- Identify operability problems and their effect on the product specifications, the debutanizer and the fuel-gas and flare systems.
- Judge the existing safeguards and rank the risk with the example matrix.
- Raise recommendations and pass the high-ranked scenarios to LOPA.
- Check selected deviations dynamically on the simulator model (Section 5.3).

2 Scope

Boundaries: scope from the battery limit at the V-201 inlet, through P-201A/B, FV-201, E-201, C-201 with E-203, E-202, V-202, P-202A/B, to the battery limits at the propane rundown (LV-203), the bottoms to the debutanizer (LV-202), the vent to fuel gas (PV-210) and the relief to flare (PSV-201); the SIS (XV-201, XV-202, XV-209) and the utilities as they enter the unit. Excluded: the upstream unit, the debutanizer, the fuel-gas, flare, steam, cooling-water and air systems themselves (their interfaces are recorded as causes and consequences). Objectives, team, risk matrix.

2.1 Nodes

Node	Description	Design intent	Design conditions	Equipment / instruments
N1	Feed surge drum V-201 and feed pumps P-201A/B	Receive about 60 m ³ /h of LPG (C3 and lighter 57 mol %) at –1 °C from the upstream unit into V-201 at 400 kPa g, hold a surge volume (12.4 m ³ of liquid at 50 %) and pump it with P-201A (P-201B steam-turbine spare) to about 2,363 kPa g.	–1 °C; 400 kPa g; pump discharge 2,363 kPa g, shut-off 2,723 kPa g	V-201 (horizontal, 24.8 m ³ , no level controller in rev A), P-201A (motor), P-201B (turbine), suction and discharge piping <i>LG-201 local gauge, TI-201, FSLL-201 on FT-201, HS-201B auto-start</i>
N2	Feed control FV-201 and preheater E-201 to tray 18	Control the feed at 60 m ³ /h (FIC-201, FV-201 fail closed) and heat it in E-201 from –1 to 75 °C with 5.82 t/h of LP steam (TIC-202, TV-202 fail closed) so it enters tray 18 about 67 % vaporized (q = 0.33).	60 m ³ /h; –1 → 75 °C; 3.50 MW; steam 5.82 t/h at 450 kPa g (156 °C)	XV-201 (SIS), FT-201, FV-201, E-201 (LP steam on the service side), line to tray 18 <i>FIC-201, FSLL-201, TIC-202, FI-206, PI-207</i>
N3	C-201 stripping section (trays 1–17), sump and reboiler E-203	Strip C3 and lighter out of the bottoms to AI-220 ≤ 2 mol % (design 1.0 %) with a boil-up of 2.24 MW from 3.79 t/h of LP steam through TV-208 (TIC-208 holds tray 8 at 76.9 °C); keep the sump level high enough for the thermosyphon to circulate.	Bottom 109 °C at 1,718 kPa g; boil-up 2.24 MW; stripping flood 74 %	C-201 trays 1–17 and sump, E-203 thermosyphon reboiler, XV-209 (SIS), TV-208, HV-209 <i>TIC-208, TI-241, TI-242, TI-243, TI-209, PDI-212, FI-205, PI-205, PI-207, LIC-202, LSLL-202, AI-220</i>

Node	Description	Design intent	Design conditions	Equipment / instruments
N4	C-201 rectifying section (trays 18–27) and reflux FV-204	Rectify the vapour to AI-219 ≤ 2.5 mol % C4 and heavier (design 1.5 %) with 49 m ³ /h of reflux at 39 °C on tray 27 (FIC-204, FV-204 fail open; TIC-227 cascade in MAN at design); top tray 45 °C.	Top 1.70 MPa g, 45 °C; reflux 49 m ³ /h; rectifying flood 77 %	C-201 trays 18–27, FV-204, reflux line from P-202A/B FIC-204, TIC-227, TI-245, TI-246, PDI-212, AI-219
N5	Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	Condense about 920 kmol/h of overhead vapour at 1.70 MPa g with 330 m ³ /h of cooling water (15 → 25.3 °C, 3.96 MW); vent the non-condensables (72.5 Nm ³ /h) to fuel gas at 400 kPa g through PV-210 so PIC-210 holds 1.70 MPa g; PSV-201 relieves to flare at 1.80 MPa g.	1.70 MPa g, 45 °C top; condensate 39 °C; CW 330 m ³ /h at 15 °C	Overhead line, E-202 (cooling water on the service side), HV-216, V-202 vapour space, PV-210, HV-210, PSV-201, flare and fuel-gas connections PIC-210, PI-211 (independent), FI-216, TI-214, TI-215, TI-213, FI-217
N6	Reflux drum V-202, pumps P-202A/B and propane product LV-203	Hold the condensate in V-202 at 50 % (7.9 m ³ of liquid at 39 °C); pump it with P-202A (P-202B spare with auto-start) at about 2,262 kPa g; 49 m ³ /h back as reflux and 32.1 m ³ /h of propane to storage through LV-203 (LIC-203, fail closed).	39 °C; 1680 kPa g drum; pumps 2,262 kPa g, shut-off 2,410 kPa g	V-202 (15.7 m ³), P-202A/B, LV-203, product line to storage LIC-203, LSHH-203, PDI-230 / PDSL-230, FI-203, AI-219, TI-213
N7	Bottoms to the debutanizer (LIC-202, LV-202)	Send 27.8 m ³ /h of C4 and heavier at 109 °C from the C-201 sump to the debutanizer at 480 kPa g on the column pressure alone (no pump), LIC-202 holding the sump at 50 % with LV-202 (fail closed).	1,718 → 480 kPa g; 109 °C; 27.8 m ³ /h	Sump outlet, LV-202, line to the debutanizer LIC-202, LSLL-202, FI-202, AI-220, TI-209
N8	Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	Supply LP steam at 450 kPa g (156 °C) to E-201 and E-203, cooling water at 15 °C to E-202, instrument air to every control valve, power to P-201A and P-202A/B; take the vent gas into the fuel-gas system and the relief into the flare.	LP steam 450 kPa g; CW 330 m ³ /h; fuel gas 400 kPa g	LP steam header, CW supply and return, instrument-air header, power, fuel-gas header, flare header PI-207, FI-216, TI-214
N9	Procedures: start-up, shutdown, emergency shutdown, maintenance	Start the column up from cold and empty, shut it down and restart it without overpressure, air ingress, cold embrittlement or LPG release; hand vessels over for entry safely.	Permissives: SIS reset, FI-216 ≥ 250 m ³ /h, sump ≥ 30 %, TIC-208 and TIC-202 in MAN ≤ 10 %, HV-209 and HV-202 open	Whole unit; SIS start permissives and trips SIS screen (XV-201, XV-202, XV-209), start permissives, ESD pushbutton

2.2 Node map

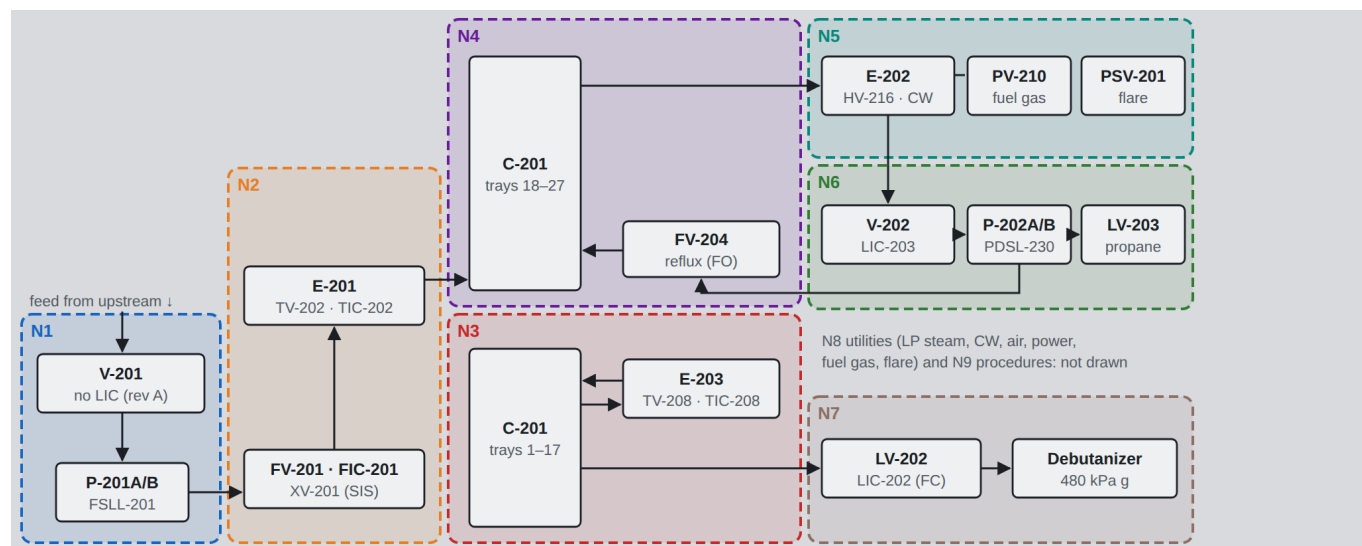


Figure 1. C-201 divided into HAZOP nodes (N8 utilities and N9 procedures cover the whole unit).

3 Methodology

HAZOP is a structured, team-based examination of a defined system. Each part of the design is compared with its **design intent**; **guide words** combined with process **parameters** generate **deviations**; for each meaningful deviation the team records causes, consequences, existing safeguards and, where the risk is not acceptable, recommendations. It finds hazards (harm to people, environment, assets) and operability problems (the column cannot make on-spec propane and bottoms, or cannot be run as intended).

3.1 References

Reference	What it contributes
IEC 61882:2016 <i>Hazard and operability studies (HAZOP studies) – Application guide</i>	Definitions, the four phases (definition, preparation, examination, documentation and follow-up), guide words, team roles, recording
CCPS <i>Guidelines for Hazard Evaluation Procedures</i>	Worksheet format, node selection, consequences judged without safeguards, risk ranking, link to LOPA
IEC 61511 (functional safety, SIS)	How HAZOP scenarios feed LOPA and SIL targets for the trips LSHH-202 and LSHH-203 and for any new high-pressure trip
ISO 31000 / owner's risk matrix	Risk criteria used to decide whether safeguards are adequate
API 520 / API 521 (ISO 23251)	Relief scenarios (cooling-water failure, blocked outlet, reflux failure, fire), PSV sizing, flare and depressuring systems
API 2510 and NFPA 58 (LPG installations)	LPG vessel relief, fire protection, water draining, spacing, BLEVE prevention
Thailand: Department of Industrial Works regulation on hazard identification, risk assessment and risk-management plans (B.E. 2543)	Legal requirement to identify hazards (HAZOP is an accepted method) and to prepare risk-management plans for factories
OSHA 29 CFR 1910.119 (PSM), where applicable	A process holding more than about 4.5 t of flammable liquid or gas is covered; this unit holds about 15 t of LPG. Process hazard analysis revalidated at least every five years

3.2 The four phases

- **Definition:** scope from the battery limit at the V-201 inlet, through P-201A/B, FV-201, E-201, C-201 with E-203, E-202, V-202, P-202A/B, to the battery limits at the propane rundown (LV-203), the bottoms to the debutanizer (LV-202), the vent to fuel gas (PV-210) and the relief to flare (PSV-201); the SIS (XV-201, XV-202, XV-209) and the utilities as they enter the unit. Excluded: the upstream unit, the debutanizer, the fuel-gas, flare, steam, cooling-water and air systems themselves (their interfaces are recorded as causes and consequences). Objectives, team, risk matrix.
- **Preparation:** collect the P&ID, PFD, heat and material balance, trip (cause-and-effect) list, alarm list, relief list, procedures and datasheets; divide the unit into nodes; write each node's design intent; choose parameters and guide words.
- **Examination:** node → design intent → parameter → guide word → is the deviation meaningful? → causes → consequences assuming the safeguards fail → existing safeguards → rank the risk → recommend → next guide word. Every meaningful deviation is recorded, including those with adequate safeguards (**full recording**), so the reasoning can be audited.
- **Documentation and follow-up:** worksheets, recommendation register with owners and dates, close-out, revalidation after changes (MOC) or periodically.

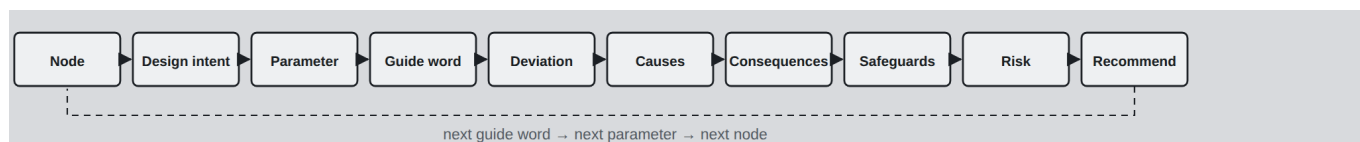


Figure 2. HAZOP examination loop.

3.3 Guide words and parameters

Guide word	Meaning (IEC 61882)
NO / NOT	Complete negation of the intent
MORE	Quantitative increase
LESS	Quantitative decrease
AS WELL AS	Qualitative increase: something extra present
PART OF	Qualitative decrease: only part of the intent achieved
REVERSE	Logical opposite of the intent

Guide word	Meaning (IEC 61882)
OTHER THAN	Complete substitution: something else happens
EARLY	Relative to clock time
LATE	Relative to clock time
BEFORE	Relative to order or sequence
AFTER	Relative to order or sequence

Parameters used: flow, level, pressure, temperature, heat, composition, efficiency, measurement, control, integrity, relief and utility; for the procedure node also sequence, purging, heating, depressuring, water draining, ESD, maintenance, safety system and steam-out. A deviation is recorded only when it is physically meaningful ("REVERSE temperature" is not).

3.4 Risk ranking

S \ L	L1	L2	L3	L4	L5
S5	5 (1)	10 (8)	15 (1)	20	25
S4	4 (1)	8 (19)	12 (3)	16	20
S3	3 (1)	6 (3)	9 (14)	12 (1)	15
S2	2	4 (6)	6 (12)	8 (5)	10
S1	1	2	3 (3)	4	5

Numbers in brackets: how many deviations of this study fall in the cell. Risk $R = S \times L$.

S	Severity (worst credible consequence)
1	First aid; minor upset; no release
2	Medical treatment; off-spec product; small contained release
3	Lost-time injury; unit shutdown for days; relief to flare; small LPG release
4	Permanent disability or single fatality; major equipment damage; LPG release with fire
5	Multiple fatalities; vessel rupture, BLEVE or vapour-cloud explosion

L	Likelihood with existing safeguards
1	Remote: not expected in the life of the plant ($< 10^{-4}$ /y)
2	Unlikely: known in industry (10^{-4} - 10^{-3} /y)
3	Possible: could happen in the plant life (10^{-3} - 10^{-2} /y)
4	Likely: several times in the plant life (10^{-2} - 10^{-1} /y)
5	Frequent: yearly or more

1-4 Low acceptable · 5-9 Medium reduce if practicable (ALARP), review safeguards · 10-16 High recommendation required, consider LOPA · 20-25 Very high not acceptable, act before operation

4 HAZOP sessions

4.1 Schedule

The study is run in class as a teaching HAZOP. A typical plan is three sessions; the team records the dates and attendance on this page.

Session	Nodes	Date	Remark
1	N1 – N3 (feed drum and pumps, feed control and preheater, stripping section, sump and reboiler)		
2	N4 – N6 (rectifying section and reflux, overhead and relief, reflux drum and propane product)		
3	N7 – N9 (bottoms, utilities, procedures); review of recommendations		

4.2 Team

Role	Responsibility	Name	Attended sessions
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live, reads back recommendations		
Process engineer (designer)	Design intent, heat and material balance, column hydraulics, relief cases and flare loads		
Operations representative	How the column is really run: rounds on LG-201, water draining, pump change-overs, start-up and shutdown, what the operators trust		
Instrument / control engineer	Loops and fail positions, the SIS (LSLL-202, LSHH-203, ESD), auto-starts, transmitter independence		
Safety / risk engineer	Risk matrix, LOPA follow-up, fire and gas, regulations		
Specialists as needed	Mechanical (pump seals, vessel design and MDMT), relief and flare, the upstream unit, the debutanizer and the utilities owners		

4.3 Documents reviewed

Document	Title / content
P&ID-001 rev C	C-201 P&ID, screen drawing of the simulator (Appendix 2)
DC3-V1-PID-001 rev D	C-201 P&ID, alternative EPC-standard drawing (Appendix 2)
DC-PFD-200	Process flow diagram with the stream table
DC3-V1-IN-CE-001 rev R0	Cause and effect diagram of the SIS
DC3-V1-IN-SRS-001 rev R0	Safety requirements specification of the SIFs
Simulator tabs	Process, Design, Control loop design, alarm and trip list (Workbook tab), start-up and shutdown procedures (SOP tab)

4.4 Nodes reviewed and guide-word coverage

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows	Flow / level	Pressure	Temperature / heat
N1	1	3	2	1	·	1	1	·	·	·	·	9	yes	yes	yes
N2	1	3	2	·	·	·	2	·	·	·	·	8	yes	yes	yes
N3	1	3	3	·	·	·	3	·	·	·	·	10	yes	yes	yes
N4	1	2	3	·	·	·	2	·	·	·	·	8	yes	yes	yes
N5	·	5	·	1	·	·	3	·	·	·	·	9	yes	yes	yes
N6	·	3	2	1	·	1	2	·	·	·	·	9	yes	yes	yes
N7	1	3	1	·	·	1	2	·	·	·	·	8	yes	yes	yes
N8	3	3	1	·	·	·	·	·	·	·	·	7	yes	yes	yes
N9	·	·	·	·	1	·	4	1	1	2	1	10	yes	yes	yes

A dot means the guide word gave no meaningful deviation for any parameter of the node (it was still considered). EARLY / LATE / BEFORE / AFTER apply mainly to the procedure node N9. The right-hand columns confirm that flow or level, pressure and temperature or heat were examined in the nodes.

5 Findings and recommendations

5.1 Risk profile

Node	Deviations	High	Medium	Low	Recommendations
N1 Feed surge drum V-201 and feed pumps P-201A/B	9	4	5	0	R01, R02, R03, R04, R05, R06, R07, R16
N2 Feed control FV-201 and preheater E-201 to tray 18	8	0	7	1	R09, R10
N3 C-201 stripping section (trays 1-17), sump and reboiler E-203	10	1	8	1	R10, R11, R12, R13, R14, R15
N4 C-201 rectifying section (trays 18-27) and reflux FV-204	8	1	4	3	R11, R16, R17, R18, R19
N5 Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	9	3	6	0	R02, R11, R20, R21, R22, R23, R24, R25, R26
N6 Reflux drum V-202, pumps P-202A/B and propane product LV-203	9	1	7	1	R02, R04, R05, R07, R16, R18, R21, R27, R28
N7 Bottoms to the debutanizer (LIC-202, LV-202)	8	0	5	3	R04, R13, R14, R18
N8 Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	7	1	6	0	R20, R23, R29, R30, R31
N9 Procedures: start-up, shutdown, emergency shutdown, maintenance	10	2	6	2	R06, R07, R32, R33, R34, R35
Total	78	13	54	11	36

High-ranked deviations

Ref	Deviation	Consequence	SxL	Rec.
N1.1	Level LESS: Low level in V-201	Level falls to the pump suction: the suction vortices (below about 10 % in the model), gas enters the casing and the pump gas-binds: no feed to C-201, pump damage, seal failure and LPG leak. With the supply lost completely the model drum reaches the vortex level in about 11 min	3x4 = 12	R01, R03
N1.2	Level MORE: V-201 overfills	Drum goes liquid-full: the model reaches 95 % about 12 min after both feed pumps stop, with no alarm in rev A. Upstream unit backs up; liquid-full drum overpressured by thermal expansion	4x3 = 12	R01, R02
N1.6	Pressure MORE: V-201 blocked in and heated	Vapour pressure rises with temperature (1,037 kPa g at 40 °C in the model); in a fire the wall above the liquid overheats: vessel rupture and BLEVE with fireball	5x2 = 10	R02, R36
N1.9	Integrity OTHER THAN: Pump seal or flange leak	LPG release at 2,363 kPa g: about 34 % flashes at once (model estimate), vapour cloud, fire or vapour-cloud explosion	5x2 = 10	R08, R36
N3.9	Integrity OTHER THAN: Leak of hot bottoms	C4 and heavier at 109 °C and 1,718 kPa g, above its atmospheric boiling point: almost all flashes, jet fire or vapour-cloud explosion	5x2 = 10	R08, R36
N4.2	Flow MORE: Too much reflux	Flooding: FIC-204 to 89 m³/h, PDI-212 to 112 kPa; TIC-208 adds steam to hold tray 8, pressure rises and PSV-201 lifts at 03:02 (up to about 2,210 Nm³/h to flare); AI-220 to 29 %	4x3 = 12	R11, R17, R36
N5.1	Pressure MORE: Overpressure: cooling water lost	Nothing condenses: pressure climbs, PSV-201 lifts at 00:27 and the model peaks at 1.868 MPa g relieving up to about 19,900 Nm³/h (6,500 Nm³ in 30 min) with no trip; V-202 drains (LIC-203 LO at 04:20, down to 0.3 %) and the reflux pumps run dry	5x3 = 15	R11, R20, R21, R36
N5.2	Pressure MORE: Overpressure: vent blocked	Non-condensables accumulate and blanket E-202: pressure rises slowly (model, PV-210 closed: PIC-210 HI at 06:19, PI-211 HI at 07:37, PSV-201 lifts at 07:41, about 330 Nm³/h)	4x3 = 12	R11, R22, R36
N5.8	Temperature MORE: External fire at C-201 or V-202	Vapour generation beyond the process cases; wall overheating above the liquid, rupture and BLEVE	5x2 = 10	R02, R08
N6.9	Integrity OTHER THAN: P-202A/B seal or flange leak	LPG at 2,262 kPa g and 39 °C: about 75 % flashes at once (model estimate), vapour cloud, fire or explosion	5x2 = 10	R08, R36

Ref	Deviation	Consequence	S×L	Rec.
N8.2	Utility NO: Cooling water lost (site-wide)	See N5: relief of C-201 to flare, and of every other column on the site at the same time	5×2 = 10	R20
N9.3	Purging PART OF: Air left in the system at start-up	LPG and air form a flammable mixture inside C-201, V-202 or the flare line: internal explosion	5×2 = 10	R32, R36
N9.6	Water draining OTHER THAN: Drain valve freezes open while draining water	Uncontrolled LPG release at the drain, vapour cloud, fire; historic LPG disasters started this way	5×2 = 10	R07, R36

5.2 Recommendations

No.	Recommendation	Node	Raised from	Action by
R01	Put V-201 level on the DCS (rev B LT-201 with LAL/LAH-201) and add a low-low level trip that stops P-201A/B before the suction vortexes (about 10 % in the model); consider the rev C cascade LIC-201 → FIC-201 so the drum level sets the feed.	N1	N1.1, N1.2, N2.1	Instrument / Process
R02	Relief and fire review: confirm the design pressures of V-201, V-202 and the E-201 process side (not on the P&ID); give V-201 its own PSV sized for fire, and confirm V-202 is protected by PSV-201 with no isolation between; add fireproofing, water deluge and a depressuring route for the LPG vessels (BLEVE prevention).	N1/N5/N6	N1.2, N1.6, N5.8, N6.7	Process / Safety
R03	Bring FSLL-201 to the console as an alarm (today it only starts P-201B) and agree with the upstream unit how a feed stop is announced in both directions.	N1	N1.1, N1.3, N2.3	Instrument / Operations
R04	Put the check valves on the P-201A/B and P-202A/B discharges into the inspection plan (a passing check valve recirculates through the idle pump); add a check valve on the bottoms line downstream of LV-202 (recommended addition).	N1/N6/N7	N1.4, N6.5, N7.8	Mechanical
R05	Provide minimum-flow recirculation lines with restriction orifices from the P-201A/B and P-202A/B discharges back to V-201 and V-202, so a closed FV-201/XV-201, or FV-204 with LV-203, cannot deadhead the running pump.	N1/N6	N1.5, N2.3, N6.4	Process
R06	Check the minimum design metal temperature of V-201, V-202, C-201 and the LPG lines against auto-refrigeration: released or depressured LPG flashes toward its atmospheric boiling point (about –42 °C for propane). Depressure slowly, never re-pressure a cold vessel, add a low-temperature alarm on TI-201.	N1/N9	N1.7, N9.5	Mechanical / Process
R07	Water draining of the LPG drums: water boots with interface level on V-201 and V-202 (recommended addition), double-block drains with a spring-return valve, drains always attended; hydrate and ice awareness at –1 °C.	N1/N6/N9	N1.8, N6.8, N9.6	Process / Operations
R08	Gas detection around P-201A/B, P-202A/B, V-201, V-202, E-203 and the bottoms line, with remote-operated shut-off valves on the V-201 and V-202 outlets and on the column bottoms; double mechanical seals with seal-leak alarms on the LPG pumps.	all	N1.9, N3.9, N5.8, N6.2, N6.9, N9.7	Instrument / Mechanical
R09	Thermal relief on the E-201 process side (recommended addition), or a locked procedure: steam off (XV-202, TV-202) before the feed is blocked in.	N2	N2.6	Process
R10	Hydrocarbon detection on the E-201 and E-203 steam condensate: the process side is at higher pressure than the steam side, so a tube leak puts LPG into the condensate return. Route the condensate through a flash pot vented to a safe place.	N2/N3	N2.7, N3.10	Process / Instrument
R11	High-pressure trip PSHH on two or three independent transmitters (recommended addition, 2003) that closes XV-209 and XV-202 (heat off) before PSV-201 lifts; LOPA to set its SIL. Today PSV-201 is the only layer against every overpressure scenario.	N3/N4/N5	N3.1, N4.2, N5.1, N5.2, N5.3, N5.6, N9.4	Process safety / Instrument
R12	Reboiler margin: alarm on TV-208 opening above 90 % and trend PI-205 (steam chest) so fouling, low header pressure and a stuck valve are seen before the bottoms go off spec.	N3	N2.5, N3.2, N7.5, N8.3	Instrument / Operations
R13	On low sump level close the bottoms line too: LSLL-202 today closes only the steam valves, so vapour at 1,718 kPa g can blow through LV-202 into the debutanizer at 480 kPa g. Add an XV on the bottoms line (recommended addition) and check the debutanizer relief for gas blow-by.	N3/N7	N3.4, N7.2, N7.7	Process / Instrument
R14	Independent high-high sump level switch LSHH-202 (recommended addition) with alarm, and a procedure to cut feed and heat when the sump cannot be emptied.	N3/N7	N3.5, N7.1, N7.3, N7.7	Instrument
R15	Deviation alarm between TIC-208 (tray 8) and the neighbouring TI-242 (tray 4) and TI-243 (tray 12) so a drifting control transmitter is seen.	N3	N3.1, N3.8	Instrument
R16	Test the auto-start of P-201B (FSLL-201) and P-202B (PDSL-230) at a fixed interval and after maintenance; record the tests.	N1/N4/N6	N1.3, N4.1, N6.3	Operations / Instrument

No.	Recommendation	Node	Raised from	Action by
R17	Reflux too high: FIC-204 HI alarm response note and a high limit on the FIC-204 set point; review the fail-open choice of FV-204 (the model floods the column and lifts PSV-201 when it fails open).	N4	N4.2, N4.4	Process / Instrument
R18	Validate AI-219 and AI-220 against a laboratory sample every shift; alarm on analyzer fault.	N4/N6/ N7	N2.1, N2.8, N3.6, N3.7, N4.3, N4.6, N6.6, N7.5	Laboratory / Instrument
R19	Composition control: run the TIC-227 → FIC-204 cascade and TIC-208 together only after a decoupling review; operating note on which loop to put in MAN first.	N4	N4.7	Process control
R20	Flare study: loss of cooling water is common to every column on the site, so check the flare header and flare for the simultaneous relief (C-201 alone reaches about 19,900 Nm ³ /h in the model); back-pressure on PSV-201.	N5/N8	N5.1, N5.3, N6.7, N8.2, N8.7	Process safety
R21	Low-low level on V-202 (LSLL-203, recommended addition) that stops P-202A/B before they run dry.	N5/N6	N5.1, N6.2	Instrument
R22	Car-seal HV-210 open (CSO register) and add a PIC-210 output-high alarm: a vent that cannot open leaves the column to PSV-201.	N5	N5.2	Operations / Instrument
R23	Fuel-gas interface: knock-out drum on the PV-210 vent before the fuel-gas header, high-flow alarm on FI-217, and tell the fuel-gas system owner that PV-210 fails open.	N5/N8	N5.4, N5.5, N6.1, N8.6	Process
R24	Deviation alarm between PIC-210 and PI-211: two independent transmitters must agree; operating note on which to believe.	N5	N5.6	Instrument
R25	PSV-201 inlet and outlet isolation valves car-sealed open; test and inspection interval; flare line free-draining.	N5	N5.7	Mechanical / Operations
R26	Hydrocarbon detection on the E-202 cooling-water return: LPG leaking into the cooling water is released at the cooling tower.	N5	N5.9, N6.8	Instrument
R27	Let LSHH-203 also close PV-210 (or confirm the knock-out drum of R23) so liquid LPG cannot reach the fuel-gas header when V-202 overfills.	N6	N6.1	Instrument / Process
R28	Off-spec propane: divert the product to an off-spec tank on AI-219 HI instead of the sales tank.	N6	N4.6, N6.6	Process / Operations
R29	Instrument-air failure: review the fail positions (PV-210 fails open to fuel gas, LV-202 and LV-203 fail closed) and consider an ESD on low air-header pressure; confirm the air receiver hold-up time.	N8	N5.5, N8.1	Instrument / Process
R30	Power failure: supply P-202B from a different bus (or a steam driver like P-201B) and write the power-failure response procedure.	N8	N8.5	Electrical / Operations
R31	Confirm the steam-side design pressure of E-201 and E-203 against the highest LP header pressure, and their relief.	N8	N8.4	Mechanical
R32	Start-up: purge C-201, V-202, V-201 and the lines with nitrogen to an oxygen content below the limit before hydrocarbon is admitted; record the gas tests; hold point in the procedure.	N9	N9.3	Operations
R33	Turnaround entry: positive isolation (blinds), nitrogen disconnected, oxygen and hydrocarbon tests, standby person and rescue plan for every vessel entry.	N9	N9.8	Operations / Safety
R34	Trip-bypass management for LSLL-202, LSHH-203 and the auto-starts: permit, time limit, bypass shown on the DCS, proof tests at the interval set by LOPA.	N9	N9.9	Operations / Instrument
R35	Steam-out: confirm the vacuum rating of C-201 and V-202, or keep a vent open while the steamed vessel cools.	N9	N9.10	Mechanical / Operations
R36	LOPA for the High rows that rely on PSV-201, LSHH-203, LSLL-202 and on the operator (overpressure from cooling-water loss, a blocked vent or too much reflux; V-201 overfill and fire; LPG leaks).	all	N1.6, N1.9, N3.9, N4.1, N4.2, N5.1, N5.2, N5.7, N6.1, N6.9, N7.2, N8.1, N9.3, N9.6	Process safety

5.3 Dynamic check of selected deviations on the simulator

A conventional HAZOP judges consequences qualitatively. Here ten deviations are run on copies of the depropanizer model from the design point with the existing safeguards active (alarms, the SIS trips, the auto-starts, PSV-201, controllers in their normal modes, no operator action) for 30 min, or 60 s past a trip. Alarms are evaluated from the page's alarm list every second with the console's suppression (product and loop alarms only while the column is heated and fed; rev B alarms only with rev B). The runs show whether the safeguards act before harm, and how much time the operator has.

10 runs finished.

Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trip (first-out) / relief	Range reached (to 60 s after a trip, or in 30 min)	Also seen
N1	Flow NO	Running feed pump P-201A trips and spare P-201B will not start	TIC-202 HI at 00:02 TIC-208 HI at 00:46 PIC-210 LO at 00:52	no trip	PI-211 1.325 to 1.700 MPa g V-201 level (LG-201) 50 to 98 % AI-219 0.1 to 1.5 mol %	top 1.32–1.700 MPa g V-201 up to 98 % (95 % at 11:59)
N3	Heat MORE	TIC-208 forced to MAN at 100 % (reboiler steam wide open)	TIC-208 HI at 00:53 AI-219 HI at 12:55 TIC-227 HI at 14:11	no trip	TIC-208 76.9 to 110.1 °C PI-211 1.616 to 1.736 MPa g AI-219 1.5 to 25.0 mol %	top 1.62–1.736 MPa g AI-219 up to 25.0 %
N4	Flow NO	Reflux pump P-202A trips and spare P-202B will not start	FIC-204 LO at 00:01 TIC-208 HI at 00:44 TIC-227 HI at 01:01 PIC-210 LO at 01:35	LSHH-203 reflux drum high-high at 05:07	FIC-204 0.0 to 49.0 m³/h LIC-203 50 to 100 % AI-219 1.5 to 9.6 mol %	top 1.08–1.701 MPa g V-202 up to 100 % AI-219 up to 9.6 %
N4	Flow MORE	Reflux valve FV-204 fails open	FIC-204 HI at 00:03 TIC-208 LO at 01:06 PIC-210 HI at 02:12 PI-211 HI at 02:59	no trip PSV-201 lifts at 03:02, up to 2,210 Nm³/h	FIC-204 49.0 to 88.9 m³/h PDI-212 18 to 112 kPa PI-211 1.699 to 1.800 MPa g	top 1.70–1.800 MPa g AI-220 up to 29.4 % 725 Nm³ to flare
N5	Pressure MORE	Cooling water to E-202 lost	FI-216 LO at 00:02 PIC-210 HI at 00:22 PI-211 HI at 00:27 TIC-208 LO at 03:09	no trip PSV-201 lifts at 00:27, up to 19,853 Nm³/h	PI-211 1.700 to 1.868 MPa g LIC-203 0 to 50 % FIC-204 0.0 to 49.4 m³/h	top 1.70–1.868 MPa g V-202 down to 0.3 % AI-220 up to 4.8 % 6,479 Nm³ to flare
N5	Pressure MORE	Vent PV-210 closed (PIC-210 forced to MAN at 0 %)	PIC-210 HI at 06:20 PI-211 HI at 07:37 TIC-208 LO at 11:29 AI-220 HI at 12:43	no trip PSV-201 lifts at 07:41, up to 326 Nm³/h	PI-211 1.700 to 1.786 MPa g FI-217 0 to 72 Nm³/h AI-220 1.0 to 2.3 mol %	top 1.70–1.786 MPa g AI-220 up to 2.3 % 25 Nm³ to flare
N6	Level MORE	Propane product valve LV-203 closed (LIC-203 forced to MAN at 0 %)	LIC-203 HI at 09:00	LSHH-203 reflux drum high-high at 13:42	LIC-203 50 to 97 % PI-211 1.516 to 1.702 MPa g FI-203 0.0 to 32.1 m³/h	top 1.52–1.702 MPa g V-202 up to 97 %
N7	Flow NO	Bottoms valve LV-202 fails closed	LIC-202 HI at 04:21 PDI-212 HI at 19:57	no trip	C-201 sump (LSLL-202) 50 to 100 % PDI-212 18 to 41 kPa FI-202 0.0 to 27.8 m³/h	top 1.70–1.700 MPa g sump up to 100 %
N8	Utility NO	Instrument air failure: every control valve to its fail position	FIC-204 HI at 00:03 TIC-202 HI at 00:04 TIC-208 LO at 00:24 PIC-210 LO at 00:31	no trip	PI-211 0.836 to 1.700 MPa g C-201 sump (LSLL-202) 50 to 100 % LIC-203 0 to 50 %	top 0.84–1.700 MPa g V-202 down to 0.3 % sump up to 100 % V-201 up to 98 % (95 % at 12:01) AI-220 up to 55.3 %
N8	Utility LESS	LP steam header falls from 450 toward 180 kPa g	PI-207 LO at 00:45 TIC-208 LO at 01:26 TIC-202 LO at 01:42 PIC-210 LO at 02:41	no trip	PI-207 180 to 450 kPa g TIC-208 57.1 to 76.9 °C AI-220 1.0 to 23.3 mol %	top 1.58–1.710 MPa g AI-220 up to 23.3 %

5.4 Follow-up and close-out

- Each recommendation gets an owner and a due date; the owner answers *accepted and done*, *accepted with an alternative* (with justification) or *rejected* (with justification agreed by the HAZOP leader and the plant manager).
- High scenarios whose protection relies on PSV-201, on LSHH-203 or LSLL-202, or on the operator (overpressure from cooling-water loss, a blocked vent or too much reflux; V-201 overfill and fire; LPG leaks) go to **LOPA** to decide whether a new high-pressure trip is needed and what SIL it must meet (IEC 61511).
- The relief recommendations (fire case on V-201 and V-202, simultaneous cooling-water failure on the flare) go to a **relief and flare study** (API 521).
- Any change (rev B LT-201 or rev C LIC-201 in service, a new trip or a changed set point, a different fail position) goes through **management of change**, which checks whether the HAZOP must be updated.
- Interfaces: loss of feed and V-201 overfill to the upstream unit; gas blow-by and off-spec bottoms to the debutanizer HAZOP; vent flow and liquid carry-over to the fuel-gas system owner; simultaneous relief to the flare owner.
- Revalidate periodically (typically every five years) and after incidents.
- Use the simulator to train operators on the scenarios the HAZOP found, and to check that the procedures and alarm responses really work (section 9).

Appendix 1 · HAZOP worksheets

Node N1 · Feed surge drum V-201 and feed pumps P-201A/B

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N1 · Feed surge drum V-201 and feed pumps P-201A/B
Design intent	Receive about 60 m ³ /h of LPG (C3 and lighter 57 mol %) at –1 °C from the upstream unit into V-201 at 400 kPa g, hold a surge volume (12.4 m ³ of liquid at 50 %) and pump it with P-201A (P-201B steam-turbine spare) to about 2,363 kPa g.	Design conditions	–1 °C; 400 kPa g; pump discharge 2,363 kPa g, shut-off 2,723 kPa g
Equipment	V-201 (horizontal, 24.8 m ³ , no level controller in rev A), P-201A (motor), P-201B (turbine), suction and discharge piping	Instruments	LG-201 local gauge, TI-201, FSLL-201 on FT-201, HS-201B auto-start
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Level	LESS	Low level in V-201	- Supply from the upstream unit lost or cut back hard - drain left open - any lasting difference between supply and feed rate (V-201 has no level controller in rev A, so the mismatch accumulates)	Level falls to the pump suction: the suction vortices (below about 10 % in the model), gas enters the casing and the pump gas-binds: no feed to C-201, pump damage, seal failure and LPG leak. With the supply lost completely the model drum reaches the vortex level in about 11 min	- LG-201 (field only) - FSLL-201 (< 15 m³/h, 5 s) auto-starts P-201B - operator rounds	3	4	12 H	R01, R03	Instrument / Operations · Instrument / Process
N1.2	Level	MORE	V-201 overfills	Feed to C-201 stopped (both pumps lost, FV-201 or XV-201 closed, instrument-air failure) while the upstream unit keeps sending	Drum goes liquid-full: the model reaches 95 % about 12 min after both feed pumps stop, with no alarm in rev A. Upstream unit backs up; liquid-full drum overpressured by thermal expansion	- LG-201 (field only) - none on the DCS in rev A	4	3	12 H	R01, R02	Instrument / Process · Process / Safety
N1.3	Flow	NO	No feed from the pumps	- P-201A trips - P-201B turbine will not start (governor, steam) - suction valve closed	No feed to C-201: the column runs down to total reflux, pressure falls (1.32 MPa g in the model) and product flows stop; V-201 fills (row above)	- FSLL-201 (< 15 m³/h, 5 s) auto-starts P-201B - TIC-202 HI alarm (85 °C) (the first alarm in the model, at 00:02) - P-201B spare	3	3	9 M	R03, R16	Instrument / Operations · Operations / Instrument
N1.4	Flow	REVERSE	Recirculation through the idle pump	Check valve of the idle pump passing	Part of the discharge returns to V-201 through the idle pump, less feed reaches C-201, the idle pump spins backwards	- Check valves - FIC-201 opens FV-201 to hold the flow	2	3	6 M	R04	Mechanical
N1.5	Pressure	MORE	Feed pump deadheaded	- FV-201 or XV-201 closed with the pump running - FV-201 fails closed	Pump at shut-off (2,723 kPa g) with no flow: LPG heats and flashes in the casing, seal failure, LPG release	None (no minimum-flow line on the P&ID)	3	3	9 M	R05	Process
N1.6	Pressure	MORE	V-201 blocked in and heated	Sun or external fire on an isolated drum	Vapour pressure rises with temperature (1,037 kPa g at 40 °C in the model); in a fire the wall above the liquid overheats: vessel rupture and BLEVE with fireball	None on the P&ID (no relief valve on V-201)	5	2	10 H	R02, R36	Process / Safety · Process safety

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.7	Temperature	LESS	Auto-refrigeration on depressuring	V-201 or a feed line depressured or leaking to atmosphere	LPG flashes toward its atmospheric boiling point (about –42 °C for propane): brittle fracture of carbon steel if re-pressured cold; cold burns	Operating procedures	4	2	8 M	R06	Mechanical / Process
N1.8	Composition	AS WELL AS	Free water in the feed	Carry-over from the upstream unit	Water settles in V-201: ice and hydrates at –1 °C in drains and instrument taps; draining LPG with the water, a drain valve can freeze open	None shown (no water boot on the P&ID)	4	2	8 M	R07	Process / Operations
N1.9	Integrity	OTHER THAN	Pump seal or flange leak	Seal failure (dry running, deadheading, gas-binding), gasket failure	LPG release at 2,363 kPa g: about 34 % flashes at once (model estimate), vapour cloud, fire or vapour-cloud explosion	Operator rounds	5	2	10 H	R08, R36	Instrument / Mechanical · Process safety

Node N2 · Feed control FV-201 and preheater E-201 to tray 18

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N2 · Feed control FV-201 and preheater E-201 to tray 18
Design intent	Control the feed at 60 m ³ /h (FIC-201, FV-201 fail closed) and heat it in E-201 from –1 to 75 °C with 5.82 t/h of LP steam (TIC-202, TV-202 fail closed) so it enters tray 18 about 67 % vaporized (q = 0.33).	Design conditions	60 m ³ /h; –1 → 75 °C; 3.50 MW; steam 5.82 t/h at 450 kPa g (156 °C)
Equipment	XV-201 (SIS), FT-201, FV-201, E-201 (LP steam on the service side), line to tray 18	Instruments	FIC-201, FSLL-201, TIC-202, FI-206, PI-207
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Flow	MORE	More feed than the column is designed for	- FIC-201 set point raised - FT-201 reads low - upstream pushes more	Column load above design: rectifying flooding, off-spec products, higher condenser duty; drum inventory changes (no level controller in rev A)	- PDI-212 HI alarm (30 kPa) - TIC-208 LO alarm (72 °C) - AI-220 HI alarm (2 mol %) (no feed-flow alarm in rev A)	2	3	6 M	R01, R18	Instrument / Process · Laboratory / Instrument
N2.2	Flow	LESS	Less feed	- FV-201 sticks or partly closes - pump wear	Turndown: weeping on the trays, lower efficiency; products stay near spec at more energy per tonne	FSLL-201 (< 15 m ³ /h, 5 s) auto-starts P-201B at very low flow	1	3	3 L	—	—
N2.3	Flow	NO	No feed through FV-201	- FV-201 fails closed (fault or air failure) - XV-201 closed by the SIS	No feed: see N1 (pump deadheaded, V-201 fills, column runs down)	FSLL-201 (< 15 m ³ /h, 5 s) auto-starts P-201B (starts P-201B, which cannot help)	3	3	9 M	R03, R05	Instrument / Operations · Process
N2.4	Temperature	MORE	Feed hotter, more vaporized	- TV-202 fails open or TIC-202 forced high - TT-202 reads low	More vapour into the rectifying section: pressure and flooding rise (model TV-202 wide open: feed 80 °C, pressure up to 1.729 MPa g, AI-219 2.8 %)	- TIC-202 HI alarm (85 °C) - PIC-210 HI alarm (1.76 MPa g) - AI-219 HI alarm (2.5 mol %)	2	3	6 M	—	—
N2.5	Temperature	LESS	Feed colder, subcooled	- E-201 steam restricted - TV-202 sticks - condensate backing up in E-201 - low header pressure	More liquid to the stripping section and more reboiler duty: bottoms off spec (model, steam restricted: TIC-202 LO at 00:52; feed 63 °C; AI-220 up to 5.9 %)	- TIC-202 LO alarm (68 °C) - TIC-208 adds steam - AI-220 HI alarm (2 mol %)	2	3	6 M	R12	Instrument / Operations
N2.6	Pressure	MORE	E-201 process side blocked in with steam on	Feed isolated for maintenance or by XV-201 while TV-202 stays open	Blocked-in LPG heated by 156 °C steam: pressure rises far above design, flange leak or rupture, LPG release	None on the P&ID (no thermal relief)	4	2	8 M	R09	Process
N2.7	Integrity	OTHER THAN	E-201 tube leak	Corrosion, vibration, thermal cycling	LPG at about 2,363 kPa g enters the steam side at 450 kPa g: hydrocarbon in the condensate return, fire or explosion in the condensate system or the boiler house	None (no detection on the condensate)	4	2	8 M	R10	Process / Instrument
N2.8	Composition	OTHER THAN	Feed composition changes	Upstream unit changes feedstock or operation	Lighter feed: more propane, AI-219 up to 1.9 % in the model; heavier feed: AI-220 up to 3.1 % (AI-220 HI at 14:22)	- AI-219 HI alarm (2.5 mol %) - AI-220 HI alarm (2 mol %) - TIC-208 moves the steam	2	4	8 M	R18	Laboratory / Instrument

Node N3 · C-201 stripping section (trays 1-17), sump and reboiler E-203

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N3 · C-201 stripping section (trays 1-17), sump and reboiler E-203
Design intent	Strip C3 and lighter out of the bottoms to AI-220 ≤ 2 mol % (design 1.0 %) with a boil-up of 2.24 MW from 3.79 t/h of LP steam through TV-208 (TIC-208 holds tray 8 at 76.9 °C); keep the sump level high enough for the thermosyphon to circulate.	Design conditions	Bottom 109 °C at 1,718 kPa g; boil-up 2.24 MW; stripping flood 74 %
Equipment	C-201 trays 1-17 and sump, E-203 thermosyphon reboiler, XV-209 (SIS), TV-208, HV-209	Instruments	TIC-208, TI-241, TI-242, TI-243, TI-209, PDI-212, FI-205, PI-205, PI-207, LIC-202, LSL-202, AI-220
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Heat	MORE	Too much boil-up	- TV-208 fails open or TIC-208 forced to MAN high - TT-208 reads low	Heavies carried overhead (model, TIC-208 MAN 100 %: tray 8 to 110 °C, AI-219 25 % after 30 min), vapour load and pressure up (1.736 MPa g), flooding; relief if the condenser cannot follow	- TIC-208 HI alarm (82 °C) - AI-219 HI alarm (2.5 mol %) - TIC-227 HI alarm (50 °C) - PSV-201	3	3	9 M	R11, R15	Instrument · Process safety / Instrument
N3.2	Heat	LESS	Too little boil-up	- E-203 fouling - LP header pressure low - TV-208 sticks - condensate backing up in E-203 - HV-209 partly closed	C3 left in the bottoms: AI-220 off spec to the debutanizer (model, header falling to 180 kPa g: AI-220 23 %, first alarm PI-207 LO at 00:45); weeping at low vapour	- PI-207 LO alarm (350 kPa g) - TIC-208 LO alarm (72 °C) - AI-220 HI alarm (2 mol %) - PI-205 steam chest	2	4	8 M	R12	Instrument / Operations
N3.3	Heat	NO	Reboiler steam cut	- XV-209 closed by LSL-202, LSHH-203 or ESD - spurious trip	Column cools and depressures, liquid drains to the sump, products stop (after the ESD the model settles near 1.51 MPa g)	- Designed trip action - restart procedure	2	3	6 M	—	—
N3.4	Level	LESS	Sump level low	- LV-202 sticks open - LIC-202 transmitter frozen or reading high - debutanizer pressure low	Reboiler circulation lost; then vapour at 1,718 kPa g blows through LV-202 into the debutanizer at 480 kPa g: overpressure downstream	- LIC-202 LO alarm (25 %) - LSL-202 trip (sump < 5 %, 5 s) closes XV-209 and XV-202 (closes the steam only, not LV-202) - LG-202	4	2	8 M	R13	Process / Instrument
N3.5	Level	MORE	Sump level high	- LV-202 fails closed - LIC-202 transmitter frozen or reading low - debutanizer pressure high	Liquid backs up over the reboiler return and the bottom trays: flooding, PDI-212 up, off-spec products (model, LV-202 closed: LIC-202 HI alarm (75 %) at 04:21, PDI-212 HI at 19:57, no trip)	- LIC-202 HI alarm (75 %) - PDI-212 HI alarm (30 kPa) - no high-level trip	3	3	9 M	R14	Instrument
N3.6	Pressure	MORE	Stripping section ΔP high (foaming or flooding)	- Foaming (feed contaminants, additives) - vapour load above flood	Entrainment, efficiency lost, bottoms off spec (model, foaming: PDI-212 HI at 00:42, up to 90 kPa; AI-220 7.6 %)	- PDI-212 HI alarm (30 kPa) - AI-220 HI alarm (2 mol %)	2	3	6 M	R18	Laboratory / Instrument
N3.7	Efficiency	LESS	Stages lost (tray damage)	Pressure surge, water slug or corrosion lifts or damages stripping trays	Same boil-up strips less: bottoms off spec without a ΔP rise	- AI-220 HI alarm (2 mol %) - tray temperatures	2	2	4 L	R18	Laboratory / Instrument
N3.8	Measurement	OTHER THAN	TT-208 (tray 8) reads wrong	Drift, thermowell fouling, wiring	TIC-208 moves the steam the wrong way: bottoms off spec (reads high) or heavies overhead and pressure up (reads low)	- TI-242, TI-243 nearby - AI-220 HI alarm (2 mol %) - AI-219 HI alarm (2.5 mol %)	2	3	6 M	R15	Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.9	Integrity	OTHER THAN	Leak of hot bottoms	Flange, gasket or reboiler channel leak	C4 and heavier at 109 °C and 1,718 kPa g, above its atmospheric boiling point: almost all flashes, jet fire or vapour-cloud explosion	Operator rounds	5	2	10 H	R08, R36	Instrument / Mechanical · Process safety
N3.10	Integrity	OTHER THAN	E-203 tube leak	Corrosion, vibration	Hydrocarbon at 1,718 kPa g into the steam chest (254 kPa g): LPG in the condensate return	PI-205 rise (not alarmed)	4	2	8 M	R10	Process / Instrument

Node N4 · C-201 rectifying section (trays 18-27) and reflux FV-204

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N4 · C-201 rectifying section (trays 18-27) and reflux FV-204
Design intent	Rectify the vapour to AI-219 ≤ 2.5 mol % C4 and heavier (design 1.5 %) with 49 m ³ /h of reflux at 39 °C on tray 27 (FIC-204, FV-204 fail open; TIC-227 cascade in MAN at design); top tray 45 °C.	Design conditions	Top 1.70 MPa g, 45 °C; reflux 49 m ³ /h; rectifying flood 77 %
Equipment	C-201 trays 18-27, FV-204, reflux line from P-202A/B	Instruments	FIC-204, TIC-227, TI-245, TI-246, PDI-212, AI-219
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Flow	NO	No reflux	- P-202A trips and P-202B does not start - FV-204 closed - V-202 empty (see N6)	Top tray heats, heavies go overhead (model: FIC-204 LO alarm (35 m ³ /h) at 00:01, TIC-227 HI at 01:01, AI-219 up to 9.6 %); the condensate stays in V-202, which overfills and trips LSHH-203 at 05:07	- PDSL-230 (< 150 kPa, 3 s) auto-starts P-202B - FIC-204 LO alarm (35 m³/h) - TIC-227 HI alarm (50 °C) - LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202	3	3	9 M	R16, R36	Operations / Instrument · Process safety
N4.2	Flow	MORE	Too much reflux	- FV-204 fails open (fail-open valve) - FIC-204 set point high - FT-204 reads low	Flooding: FIC-204 to 89 m ³ /h, PDI-212 to 112 kPa; TIC-208 adds steam to hold tray 8, pressure rises and PSV-201 lifts at 03:02 (up to about 2,210 Nm ³ /h to flare); AI-220 to 29 %	- FIC-204 HI alarm (65 m³/h) - PDI-212 HI alarm (30 kPa) - PI-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201	4	3	12 H	R11, R17, R36	Process / Instrument · Process safety · Process safety / Instrument
N4.3	Flow	LESS	Too little reflux	- FIC-204 set point low - FV-204 sticks - pump wear	Heavies overhead: propane product off spec	- AI-219 HI alarm (2.5 mol %) - TIC-227 HI alarm (50 °C)	2	3	6 M	R18	Laboratory / Instrument
N4.4	Pressure	MORE	Rectifying ΔP high (flooding)	- Feed too hot or too much feed - too much boil-up - reflux too high	Entrainment, efficiency lost, both products off spec; ΔP swings	PDI-212 HI alarm (30 kPa)	2	3	6 M	R17	Process / Instrument
N4.5	Flow	LESS	Vapour too low (weeping)	Deep turndown, boil-up cut	Liquid weeps through the trays, efficiency lost	- Tray temperatures - AI-219, AI-220	1	3	3 L	—	—
N4.6	Composition	OTHER THAN	Propane product off spec	- Any of the rows above - lighter or heavier feed - analyzer fault	Product downgraded (330 instead of 420 \$/m ³ in the training prices); off-spec propane into the sales tank	AI-219 HI alarm (2.5 mol %)	2	4	8 M	R18, R28	Laboratory / Instrument · Process / Operations
N4.7	Control	OTHER THAN	Two composition loops fight	TIC-227 cascade and TIC-208 both on automatic (interacting loops)	Reflux and boil-up cycle against each other; both products swing	TIC-227 normally in MAN	2	2	4 L	R19	Process control
N4.8	Temperature	LESS	Reflux subcooled	- Cold cooling water - condenser over-surfaced	More internal reflux, a little more boil-up; no safety effect	None needed	1	3	3 L	—	—

Node N5 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N5 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201
Design intent	Condense about 920 kmol/h of overhead vapour at 1.70 MPa g with 330 m ³ /h of cooling water (15 → 25.3 °C, 3.96 MW); vent the non-condensables (72.5 Nm ³ /h) to fuel gas at 400 kPa g through PV-210 so PIC-210 holds 1.70 MPa g; PSV-201 relieves to flare at 1.80 MPa g.	Design conditions	1.70 MPa g, 45 °C top; condensate 39 °C; CW 330 m ³ /h at 15 °C
Equipment	Overhead line, E-202 (cooling water on the service side), HV-216, V-202 vapour space, PV-210, HV-210, PSV-201, flare and fuel-gas connections	Instruments	PIC-210, PI-211 (independent), FI-216, TI-214, TI-215, TI-213, FI-217
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Pressure	MORE	Overpressure: cooling water lost	- CW pump or cooling-tower failure (site-wide) - HV-216 closed	Nothing condenses: pressure climbs, PSV-201 lifts at 00:27 and the model peaks at 1.868 MPa g relieving up to about 19,900 Nm ³ /h (6,500 Nm ³ in 30 min) with no trip; V-202 drains (LIC-203 LO at 04:20, down to 0.3 %) and the reflux pumps run dry	- FI-216 LO alarm (250 m³/h) (first, at 00:02) - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201 - no high-pressure trip	5	3	15 H	R11, R20, R21, R36	Instrument · Process safety · Process safety / Instrument
N5.2	Pressure	MORE	Overpressure: vent blocked	- HV-210 closed by mistake - PV-210 fails closed or PIC-210 in MAN at low output - fuel-gas header blocked	Non-condensables accumulate and blanket E-202: pressure rises slowly (model, PV-210 closed: PIC-210 HI at 06:19, PI-211 HI at 07:37, PSV-201 lifts at 07:41, about 330 Nm ³ /h)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201	4	3	12 H	R11, R22, R36	Operations / Instrument · Process safety · Process safety / Instrument
N5.3	Pressure	MORE	Overpressure: condenser capacity reduced	- Warmer cooling water - condenser fouling	Higher condensing temperature: PV-210 vents more (and more propane with it), then the pressure rises to relief (model: CW 15 °C warmer lifts PSV-201 at 01:43, up to 6,100 Nm ³ /h; condenser UA –50 % lifts it at 01:52, vent up to 521 Nm ³ /h)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - TI-214, TI-215 - PSV-201	4	2	8 M	R11, R20	Process safety · Process safety / Instrument
N5.4	Composition	AS WELL AS	More non-condensables	Methane or hydrogen from the upstream unit	Condenser area blanketed; PV-210 vents more gas and propane (model, five-fold: vent up to 361 Nm ³ /h, PIC-210 HI at 03:29, PSV-201 briefly lifts at 04:17)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter	3	3	9 M	R23	Process
N5.5	Flow	MORE	Too much vent to fuel gas (column depressured)	PV-210 fails open (instrument air) or PIC-210 forced high	Propane vapour lost to fuel gas (model, PV-210 wide open: vent up to 483 Nm ³ /h, pressure down to 1.54 MPa g, PIC-210 LO at 05:20); fuel-gas header overpressure; column upset	- PIC-210 LO alarm (1.6 MPa g) - fuel-gas system relief	3	3	9 M	R23, R29	Instrument / Process · Process
N5.6	Measurement	OTHER THAN	PIC-210 transmitter wrong	Drift, impulse-line leak or blockage, calibration error	PV-210 acts on a wrong pressure: the true column pressure can move away from 1.70 MPa g in either direction without the control alarm seeing it	- PI-211 independent transmitter with its own HI alarm - PSV-201	4	2	8 M	R24, R11	Instrument · Process safety / Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.7	Relief	OTHER THAN	PSV-201 unavailable or passing	- Inlet or outlet isolation closed - set too high - seat damage after lifting	No protection against the overpressure rows above (vessel rupture), or continuous loss of propane to flare	Relief valve test programme	5	1	5 M	R25, R36	Mechanical / Operations · Process safety
N5.8	Temperature	MORE	External fire at C-201 or V-202	Pool fire from an LPG leak at the pumps or the bottoms	Vapour generation beyond the process cases; wall overheating above the liquid, rupture and BLEVE	PSV-201 (fire case to be confirmed)	5	2	10 H	R02, R08	Instrument / Mechanical · Process / Safety
N5.9	Integrity	OTHER THAN	E-202 tube leak	Corrosion, vibration	LPG at 1.70 MPa g into the cooling water: flammable gas released at the cooling tower	None	4	2	8 M	R26	Instrument

Node N6 · Reflux drum V-202, pumps P-202A/B and propane product LV-203

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N6 · Reflux drum V-202, pumps P-202A/B and propane product LV-203
Design intent	Hold the condensate in V-202 at 50 % (7.9 m ³ of liquid at 39 °C); pump it with P-202A (P-202B spare with auto-start) at about 2,262 kPa g; 49 m ³ /h back as reflux and 32.1 m ³ /h of propane to storage through LV-203 (LIC-203, fail closed).	Design conditions	39 °C; 1680 kPa g drum; pumps 2,262 kPa g, shut-off 2,410 kPa g
Equipment	V-202 (15.7 m ³), P-202A/B, LV-203, product line to storage	Instruments	LIC-203, LSHH-203, PDI-230 / PDSL-230, FI-203, AI-219, TI-213
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Level	MORE	V-202 overfills	- LV-203 fails closed or storage blocked in - LIC-203 in MAN at low output - reflux lost	Liquid LPG rises into the vent to the fuel-gas header (burner flooding, fire at the users). Model, LV-203 closed: LIC-203 HI alarm (75 %) at 09:00, LSHH-203 trips at 13:43 (feed and heat off)	- LIC-203 HI alarm (75 %) - LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202	4	2	8 M	R27, R23, R36	Instrument / Process · Process · Process safety
N6.2	Level	LESS	V-202 empties	- LV-203 opens wide (LIC-203 transmitter reads high) - cooling water lost (nothing condenses)	Pumps lose suction and run dry: seal failure, LPG release; reflux lost (model, cooling water lost: LIC-203 LO at 04:20, level down to 0.3 %)	- LIC-203 LO alarm (25 %) - PDSL-230 starts the spare, which also runs dry - no low-low trip	3	3	9 M	R21, R08	Instrument · Instrument / Mechanical
N6.3	Pressure	LESS	Reflux pump ΔP low	- P-202A trips - pump gas-bound	Reflux and product interrupted; with P-202B starting automatically the model dips the reflux to 10 m ³ /h and recovers (alarms: FIC-204 LO 00:01)	- PDSL-230 (< 150 kPa, 3 s) auto-starts P-202B - FIC-204 LO alarm (35 m³/h)	2	3	6 M	R16	Operations / Instrument
N6.4	Pressure	MORE	Reflux pumps deadheaded	FV-204 and LV-203 both closed with a pump running	Pump at shut-off (2,410 kPa g): overheating, flashing, seal failure	None (no minimum-flow line)	3	2	6 M	R05	Process
N6.5	Flow	REVERSE	Back-flow through the idle pump	Check valve of the idle pump passing	Recirculation, less reflux and product, idle pump spins backwards	- Check valves - FIC-204 opens FV-204	2	2	4 L	R04	Mechanical
N6.6	Composition	OTHER THAN	Off-spec propane to storage	- Heavies overhead (N4) - analyzer fault	Sales tank contaminated, product downgraded	AI-219 HI alarm (2.5 mol %)	2	3	6 M	R18, R28	Laboratory / Instrument · Process / Operations
N6.7	Temperature	MORE	Reflux drum hot	- Cooling water lost or warm - condenser fouled	Propane vapour pressure rises with temperature (1,908 kPa g at 50 °C in the model, above the PSV-201 set pressure): the drum is held up at relief pressure through the column; reflux pumps lose NPSH	- TI-213 - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201 (if V-202 cannot be isolated from C-201)	3	3	9 M	R02, R20	Process / Safety · Process safety
N6.8	Composition	AS WELL AS	Water in V-202	- Water in the feed - condenser tube leak	Free water with the product; ice or hydrates on draining; drain valve can freeze open	None shown (no water boot)	4	2	8 M	R07, R26	Instrument · Process / Operations
N6.9	Integrity	OTHER THAN	P-202A/B seal or flange leak	Seal failure after dry running or deadheading	LPG at 2,262 kPa g and 39 °C: about 75 % flashes at once (model estimate), vapour cloud, fire or explosion	Operator rounds	5	2	10 H	R08, R36	Instrument / Mechanical · Process safety

Node N7 · Bottoms to the debutanizer (LIC-202, LV-202)

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N7 · Bottoms to the debutanizer (LIC-202, LV-202)
Design intent	Send 27.8 m ³ /h of C4 and heavier at 109 °C from the C-201 sump to the debutanizer at 480 kPa g on the column pressure alone (no pump), LIC-202 holding the sump at 50 % with LV-202 (fail closed).	Design conditions	1,718 → 480 kPa g; 109 °C; 27.8 m ³ /h
Equipment	Sump outlet, LV-202, line to the debutanizer	Instruments	LIC-202, LSSL-202, FI-202, AI-220, TI-209
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Flow	NO	No bottoms to the debutanizer	- LV-202 fails closed (fault or instrument air) - block valve closed	Debutanizer loses its feed; the C-201 sump fills (model: LIC-202 HI at 04:21, sump full, PDI-212 HI at 19:57, no trip)	- LIC-202 HI alarm (75 %) - PDI-212 HI alarm (30 kPa)	3	3	9 M	R14	Instrument
N7.2	Flow	MORE	Gas blow-by to the debutanizer	Sump emptied (LV-202 sticks open, LIC-202 transmitter frozen or reading high, bypass open)	Vapour at 1,718 kPa g flows into the debutanizer at 480 kPa g: overpressure of the debutanizer, relief there	- LIC-202 LO alarm (25 %) - LSSL-202 trip (sump < 5 %, 5 s) closes XV-209 and XV-202 (closes the steam, not LV-202)	4	2	8 M	R13, R36	Process / Instrument · Process safety
N7.3	Pressure	MORE	Debutanizer pressure high	Debutanizer upset (toward 1.5 MPa g)	Less driving pressure: LV-202 opens further, the sump rises (model: 70 % with no alarm)	LIC-202 HI alarm (75 %)	2	3	6 M	R14	Instrument
N7.4	Pressure	LESS	Debutanizer pressure low	Debutanizer depressured or shut down	Higher flow and flashing across LV-202, vibration; sump drawn down	LIC-202	2	2	4 L	—	—
N7.5	Composition	OTHER THAN	Propane in the bottoms	- Too little boil-up (N3) - feed change	Debutanizer overhead overloaded with C3: its pressure rises; product losses	AI-220 HI alarm (2 mol %)	2	4	8 M	R12, R18	Instrument / Operations · Laboratory / Instrument
N7.6	Temperature	MORE	Bottoms hotter than intended	- Too much boil-up - heavier feed	More flashing across LV-202 into the debutanizer feed line: vibration, debutanizer upset	TI-209 (indicated only)	2	2	4 L	—	—
N7.7	Measurement	OTHER THAN	LIC-202 transmitter frozen	Transmitter or impulse-line fault	The reading looks steady while the true level drifts: the sump can empty (blow-by) or fill (flooding) without an alarm on LIC-202	- LSSL-202 (independent low-low switch) - LG-202	4	2	8 M	R13, R14	Instrument · Process / Instrument
N7.8	Flow	REVERSE	Back-flow from the debutanizer	C-201 depressured below the debutanizer during a shutdown	Debutanizer liquid back into C-201	None shown (no check valve)	2	2	4 L	R04	Mechanical

Node N8 · Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N8 · Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare
Design intent	Supply LP steam at 450 kPa g (156 °C) to E-201 and E-203, cooling water at 15 °C to E-202, instrument air to every control valve, power to P-201A and P-202A/B; take the vent gas into the fuel-gas system and the relief into the flare.	Design conditions	LP steam 450 kPa g; CW 330 m³/h; fuel gas 400 kPa g
Equipment	LP steam header, CW supply and return, instrument-air header, power, fuel-gas header, flare header	Instruments	PI-207, FI-216, TI-214
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N8.1	Utility	NO	Instrument air lost	Compressor or dryer failure, header break	Every control valve goes to its fail position (as marked on the P&ID). Model: the column depressures to 0.83 MPa g through PV-210 into the fuel gas, the sump fills, V-202 empties to 0.3 %, V-201 goes liquid-full (95 % after about 12 min), no trip	- Fail positions - alarms on many loops at once	3	3	9 M	R29, R36	Instrument / Process · Process safety
N8.2	Utility	NO	Cooling water lost (site-wide)	Cooling-tower pumps or power lost	See N5: relief of C-201 to flare, and of every other column on the site at the same time	- FI-216 LO alarm (250 m³/h) - PSV-201	5	2	10 H	R20	Process safety
N8.3	Utility	LESS	LP steam header pressure low	Boiler or header upset (toward 180 kPa g)	Less boil-up and preheat: bottoms off spec (model: PI-207 LO at 00:45, AI-220 HI at 03:26, up to 23 %)	- PI-207 LO alarm (350 kPa g) - TIC-208 LO alarm (72 °C) - AI-220 HI alarm (2 mol %)	2	4	8 M	R12	Instrument / Operations
N8.4	Utility	MORE	LP steam header pressure high	Let-down valve fails open upstream	E-201 and E-203 steam sides above their design pressure; more heat available if TV-208/TV-202 open	Header relief (outside scope)	3	2	6 M	R31	Mechanical
N8.5	Utility	NO	Power lost	Grid or substation failure	P-201A and P-202A/B stop (P-201B is a steam turbine): reflux lost, V-202 overfills and LSHH-203 trips (as in N4 no reflux)	- LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202 - P-201B on steam	3	3	9 M	R30	Electrical / Operations
N8.6	Utility	MORE	Fuel-gas header pressure high	Fuel-gas users shut down	Less vent through PV-210: pressure rises as in a blocked vent	- PIC-210 HI alarm (1.76 MPa g) - PSV-201	3	2	6 M	R23	Process
N8.7	Utility	MORE	Flare header back-pressure high	Several units relieving together (CW loss)	PSV-201 relieves less than its rating: higher column peak pressure	Flare design (outside scope)	4	2	8 M	R20	Process safety

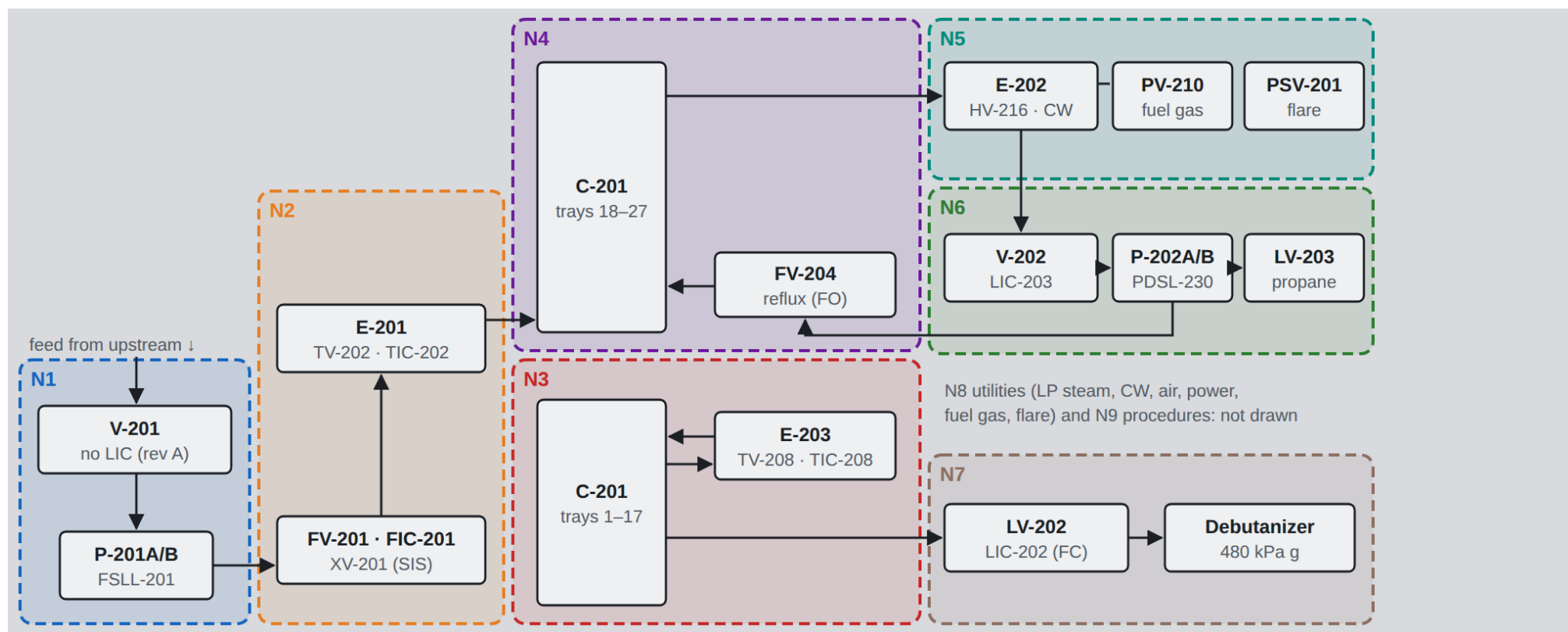
Node N9 · Procedures: start-up, shutdown, emergency shutdown, maintenance

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Node	N9 · Procedures: start-up, shutdown, emergency shutdown, maintenance
Design intent	Start the column up from cold and empty, shut it down and restart it without overpressure, air ingress, cold embrittlement or LPG release; hand vessels over for entry safely.	Design conditions	Permissives: SIS reset, FI-216 ≥ 250 m ³ /h, sump ≥ 30 %, TIC-208 and TIC-202 in MAN ≤ 10 %, HV-209 and HV-202 open
Equipment	Whole unit; SIS start permissives and trips	Instruments	SIS screen (XV-201, XV-202, XV-209), start permissives, ESD pushbutton
Drawing	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Document	DC3-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N9.1	Sequence	BEFORE	Heat before the condenser has cooling water	Wrong order at start-up	Vapour with nowhere to condense: pressure rises to relief	Start permissive FI-216 ≥ 250 m ³ /h	4	1	4 L	—	—
N9.2	Sequence	BEFORE	Reboiler steam before the sump is filled	Wrong order at start-up	Thermosyphon does not circulate; dry tubes, then a surge when liquid arrives	Start permissive: sump ≥ 30 %	3	1	3 L	—	—
N9.3	Purging	PART OF	Air left in the system at start-up	Nitrogen purge incomplete or skipped	LPG and air form a flammable mixture inside C-201, V-202 or the flare line: internal explosion	Procedure (no oxygen check point on the P&ID)	5	2	10 H	R32, R36	Operations · Process safety
N9.4	Heating	EARLY	Heat raised too fast	Steam valves opened in large steps at start-up	Pressure surge, flooding, PSV-201 lift, tray damage	- PIC-210 alarms - procedure	3	3	9 M	R11	Process safety / Instrument
N9.5	Depressuring	OTHER THAN	Liquid LPG depressured at shutdown	Vessel depressured before it is drained	Auto-refrigeration toward -42 °C: brittle fracture risk, especially if re-pressured cold	Procedure	4	2	8 M	R06	Mechanical / Process
N9.6	Water draining	OTHER THAN	Drain valve freezes open while draining water	LPG follows the water through the drain and auto-refrigerates	Uncontrolled LPG release at the drain, vapour cloud, fire; historic LPG disasters started this way	Procedure	5	2	10 H	R07, R36	Process / Operations · Process safety
N9.7	ESD	LATE	Emergency shutdown delayed	Operator unsure, many alarms at once	Leak or fire continues to be fed	ESD pushbutton (closes XV-201, XV-202, XV-209)	4	2	8 M	R08	Instrument / Mechanical
N9.8	Maintenance	OTHER THAN	Vessel entry at turnaround	- Nitrogen or hydrocarbon not cleared - isolation incomplete	Asphyxiation or fire inside the vessel	- Permit to work - gas test	4	2	8 M	R33	Operations / Safety
N9.9	Safety system	OTHER THAN	Trip bypassed	LSLL-202 or LSHH-203 bypassed for testing or nuisance trips	No protection against reboiler dry-out or liquid to fuel gas	Permit to work	4	2	8 M	R34	Operations / Instrument
N9.10	Steam-out	AFTER	Vessel closed after steaming out	Steam condenses in a closed column or drum as it cools	Vacuum: shell or tray damage if not vacuum-rated	Procedure	4	2	8 M	R35	Mechanical / Operations

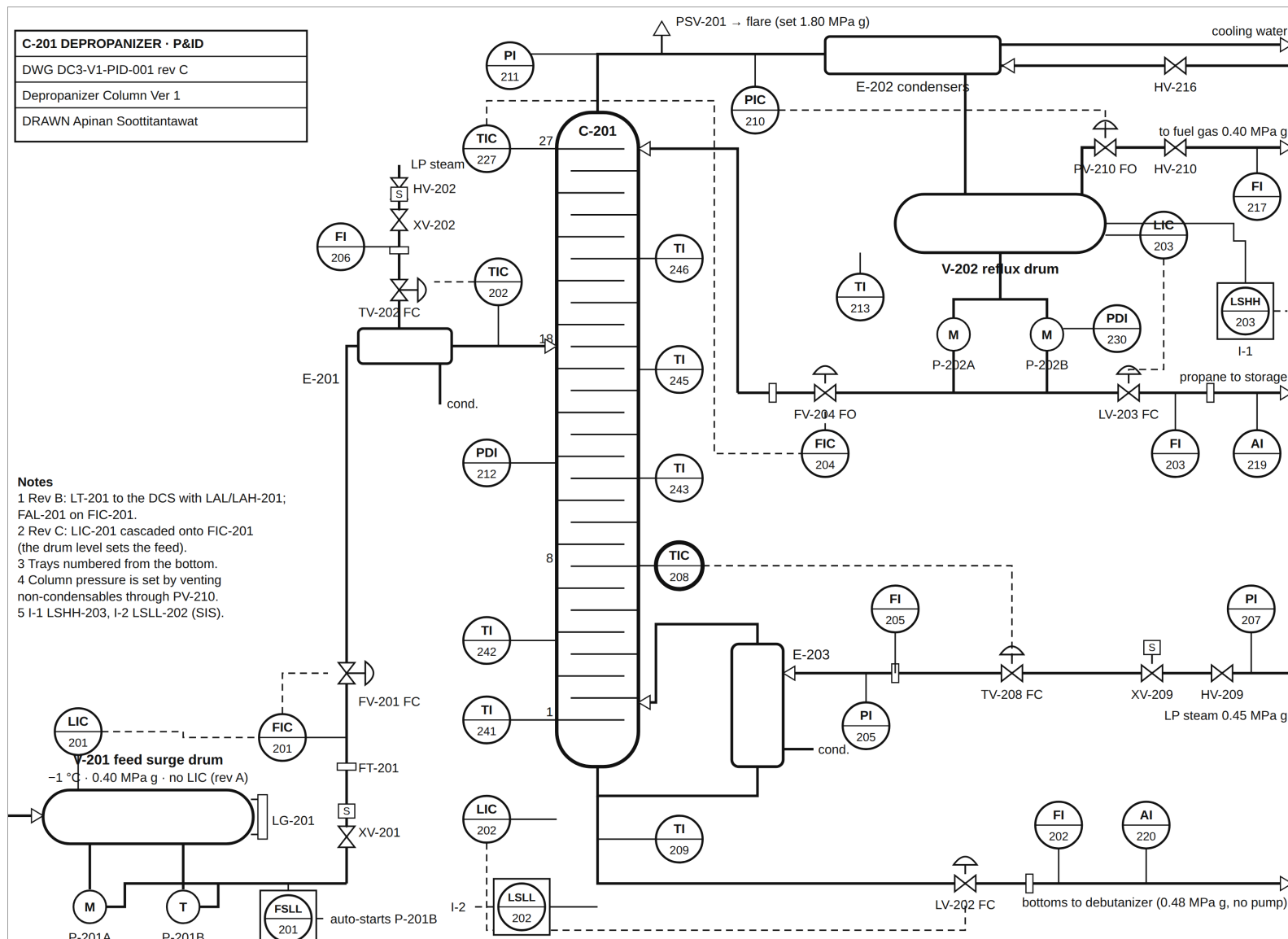
Appendix 2 · HAZOP documents (drawings with node boundaries)

A2-1 · HAZOP node map



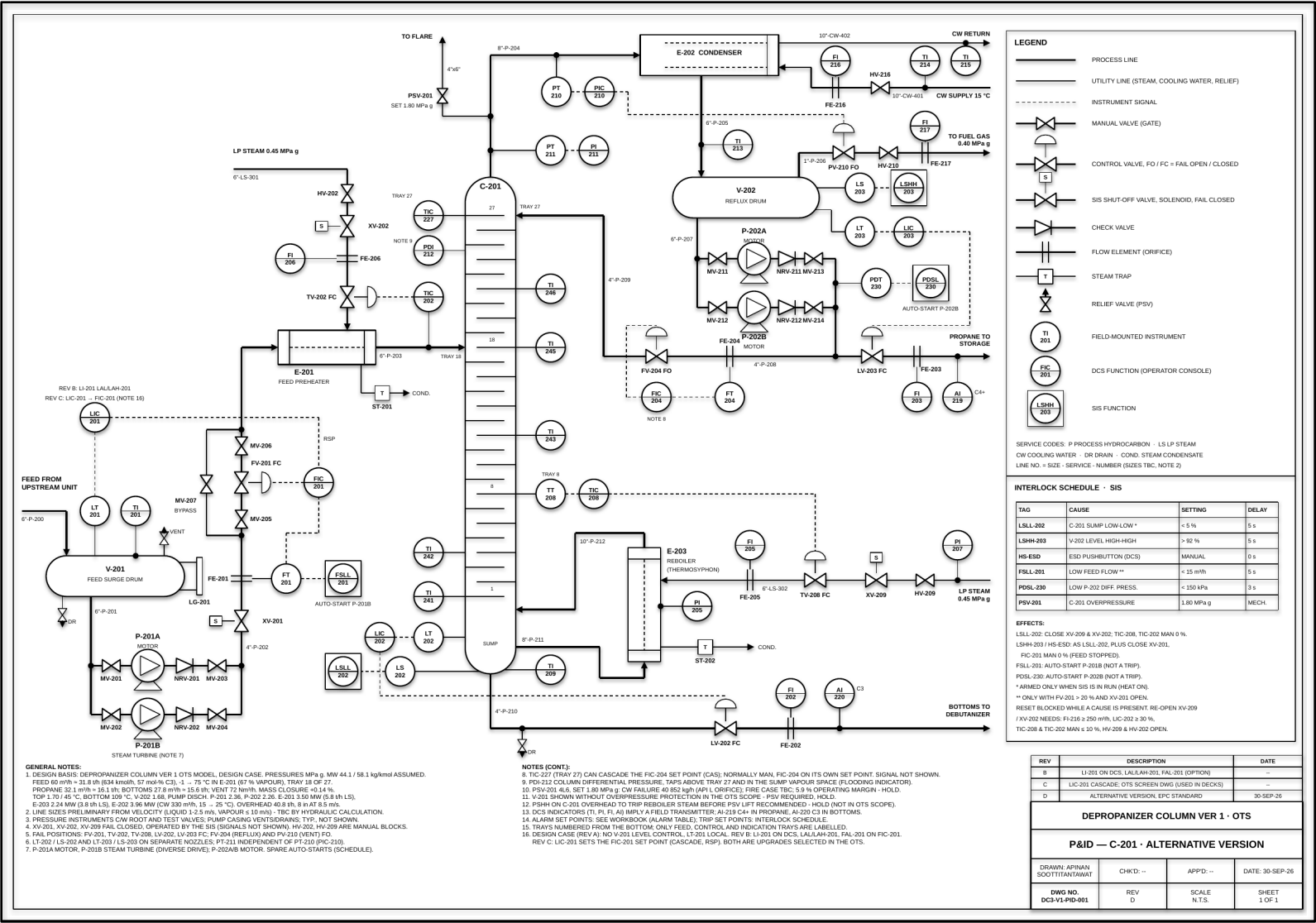
A2-1. HAZOP node map of C-201 (nodes N1–N7; N8 = utilities, N9 = procedures).

A2-2 · P&ID-001 rev C (screen drawing used for the study)



P&ID of C-201, rev C, as drawn on the P&ID tab of the simulator.

A2-3 · DC3-V1-PID-001 rev D (alternative EPC drawing)



P&ID of C-201, rev D, EPC-standard alternative version with interlock schedule.

Appendix 3 · HAZOP action close-out sheets

One sheet per recommendation (36 sheets). The action party writes the response, ticks the status and signs; the HAZOP leader accepts the close-out.

HAZOP ACTION CLOSE-OUT SHEET · R01

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R01
Node	N1 · Feed surge drum V-201 and feed pumps P-201A/B	Action party	Instrument / Process
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Level LESS : Low level in V-201	- Supply from the upstream unit lost or cut back hard - drain left open - any lasting difference between supply and feed rate (V-201 has no level controller in rev A, so the mismatch accumulates)	Level falls to the pump suction: the suction vortexes (below about 10 % in the model), gas enters the casing and the pump gas-binds: no feed to C-201, pump damage, seal failure and LPG leak. With the supply lost completely the model drum reaches the vortex level in about 11 min	- LG-201 (field only) - FSL-201 (< 15 m³/h, 5 s) auto-starts P-201B - operator rounds	S3 L4 12 High
N1.2	Level MORE : V-201 overfills	Feed to C-201 stopped (both pumps lost, FV-201 or XV-201 closed, instrument-air failure) while the upstream unit keeps sending	Drum goes liquid-full: the model reaches 95 % about 12 min after both feed pumps stop, with no alarm in rev A. Upstream unit backs up; liquid-full drum overpressured by thermal expansion	- LG-201 (field only) - none on the DCS in rev A	S4 L3 12 High
N2.1	Flow MORE : More feed than the column is designed for	- FIC-201 set point raised - FT-201 reads low - upstream pushes more	Column load above design: rectifying flooding, off-spec products, higher condenser duty; drum inventory changes (no level controller in rev A)	- PDI-212 HI alarm (30 kPa) - TIC-208 LO alarm (72 °C) - AI-220 HI alarm (2 mol %) (no feed-flow alarm in rev A)	S2 L3 6 Medium

HAZOP recommendation

Put V-201 level on the DCS (rev B LT-201 with LAL/LAH-201) and add a low-low level trip that stops P-201A/B before the suction vortexes (about 10 % in the model); consider the rev C cascade LIC-201 → FIC-201 so the drum level sets the feed.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R02

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R02
Node	N1/N5/N6 · Feed surge drum V-201 and feed pumps P-201A/B / Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201 / Reflux drum V-202, pumps P-202A/B and propane product LV-203	Action party	Process / Safety
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Level MORE : V-201 overfills	Feed to C-201 stopped (both pumps lost, FV-201 or XV-201 closed, instrument-air failure) while the upstream unit keeps sending	Drum goes liquid-full: the model reaches 95 % about 12 min after both feed pumps stop, with no alarm in rev A. Upstream unit backs up; liquid-full drum overpressured by thermal expansion	• LG-201 (field only) • none on the DCS in rev A	S4 L3 12 High
N1.6	Pressure MORE : V-201 blocked in and heated	Sun or external fire on an isolated drum	Vapour pressure rises with temperature (1,037 kPa g at 40 °C in the model); in a fire the wall above the liquid overheats: vessel rupture and BLEVE with fireball	None on the P&ID (no relief valve on V-201)	S5 L2 10 High
N5.8	Temperature MORE : External fire at C-201 or V-202	Pool fire from an LPG leak at the pumps or the bottoms	Vapour generation beyond the process cases; wall overheating above the liquid, rupture and BLEVE	PSV-201 (fire case to be confirmed)	S5 L2 10 High
N6.7	Temperature MORE : Reflux drum hot	• Cooling water lost or warm • condenser fouled	Propane vapour pressure rises with temperature (1,908 kPa g at 50 °C in the model, above the PSV-201 set pressure): the drum is held up at relief pressure through the column; reflux pumps lose NPSH	• TI-213 • PIC-210 HI alarm (1.76 MPa g) • PI-211 HI alarm (1.78 MPa g) on the independent transmitter • PSV-201 (if V-202 cannot be isolated from C-201)	S3 L3 9 Medium

HAZOP recommendation

Relief and fire review: confirm the design pressures of V-201, V-202 and the E-201 process side (not on the P&ID); give V-201 its own PSV sized for fire, and confirm V-202 is protected by PSV-201 with no isolation between; add fireproofing, water deluge and a depressuring route for the LPG vessels (BLEVE prevention).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R03

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R03
Node	N1 · Feed surge drum V-201 and feed pumps P-201A/B	Action party	Instrument / Operations
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Level LESS : Low level in V-201	- Supply from the upstream unit lost or cut back hard - drain left open - any lasting difference between supply and feed rate (V-201 has no level controller in rev A, so the mismatch accumulates)	Level falls to the pump suction: the suction vortexes (below about 10 % in the model), gas enters the casing and the pump gas-binds: no feed to C-201, pump damage, seal failure and LPG leak. With the supply lost completely the model drum reaches the vortex level in about 11 min	- LG-201 (field only) - FSSL-201 (< 15 m³/h, 5 s) auto-starts P-201B - operator rounds	S3 L4 12 High
N1.3	Flow NO : No feed from the pumps	- P-201A trips - P-201B turbine will not start (governor, steam) - suction valve closed	No feed to C-201: the column runs down to total reflux, pressure falls (1.32 MPa g in the model) and product flows stop; V-201 fills (row above)	- FSSL-201 (< 15 m³/h, 5 s) auto-starts P-201B - TIC-202 HI alarm (85 °C) (the first alarm in the model, at 00:02) - P-201B spare	S3 L3 9 Medium
N2.3	Flow NO : No feed through FV-201	- FV-201 fails closed (fault or air failure) - XV-201 closed by the SIS	No feed: see N1 (pump deadheaded, V-201 fills, column runs down)	FSSL-201 (< 15 m³/h, 5 s) auto-starts P-201B (starts P-201B, which cannot help)	S3 L3 9 Medium

HAZOP recommendation

Bring FSSL-201 to the console as an alarm (today it only starts P-201B) and agree with the upstream unit how a feed stop is announced in both directions.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R04

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R04
Node	N1/N6/N7 · Feed surge drum V-201 and feed pumps P-201A/B / Reflux drum V-202, pumps P-202A/B and propane product LV-203 / Bottoms to the debutanizer (LIC-202, LV-202)	Action party	Mechanical
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.4	Flow REVERSE : Recirculation through the idle pump	Check valve of the idle pump passing	Part of the discharge returns to V-201 through the idle pump, less feed reaches C-201, the idle pump spins backwards	- Check valves - FIC-201 opens FV-201 to hold the flow	S2 L3 6 Medium
N6.5	Flow REVERSE : Back-flow through the idle pump	Check valve of the idle pump passing	Recirculation, less reflux and product, idle pump spins backwards	- Check valves - FIC-204 opens FV-204	S2 L2 4 Low
N7.8	Flow REVERSE : Back-flow from the debutanizer	C-201 depressured below the debutanizer during a shutdown	Debutanizer liquid back into C-201	None shown (no check valve)	S2 L2 4 Low

HAZOP recommendation

Put the check valves on the P-201A/B and P-202A/B discharges into the inspection plan (a passing check valve recirculates through the idle pump); add a check valve on the bottoms line downstream of LV-202 (recommended addition).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R05

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R05
Node	N1/N6 · Feed surge drum V-201 and feed pumps P-201A/B / Reflux drum V-202, pumps P-202A/B and propane product LV-203	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.5	Pressure MORE : Feed pump deadheaded	- FV-201 or XV-201 closed with the pump running - FV-201 fails closed	Pump at shut-off (2,723 kPa g) with no flow: LPG heats and flashes in the casing, seal failure, LPG release	None (no minimum-flow line on the P&ID)	S3 L3 9 Medium
N2.3	Flow NO : No feed through FV-201	- FV-201 fails closed (fault or air failure) - XV-201 closed by the SIS	No feed: see N1 (pump deadheaded, V-201 fills, column runs down)	FSL-201 (< 15 m³/h, 5 s) auto-starts P-201B (starts P-201B, which cannot help)	S3 L3 9 Medium
N6.4	Pressure MORE : Reflux pumps deadheaded	FV-204 and LV-203 both closed with a pump running	Pump at shut-off (2,410 kPa g): overheating, flashing, seal failure	None (no minimum-flow line)	S3 L2 6 Medium

HAZOP recommendation

Provide minimum-flow recirculation lines with restriction orifices from the P-201A/B and P-202A/B discharges back to V-201 and V-202, so a closed FV-201/XV-201, or FV-204 with LV-203, cannot deadhead the running pump.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R06

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R06
Node	N1/N9 · Feed surge drum V-201 and feed pumps P-201A/B / Procedures: start-up, shutdown, emergency shutdown, maintenance	Action party	Mechanical / Process
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.7	Temperature LESS : Auto-refrigeration on depressuring	V-201 or a feed line depressured or leaking to atmosphere	LPG flashes toward its atmospheric boiling point (about –42 °C for propane): brittle fracture of carbon steel if re-pressured cold; cold burns	Operating procedures	S4 L2 8 Medium
N9.5	Depressuring OTHER THAN : Liquid LPG depressured at shutdown	Vessel depressured before it is drained	Auto-refrigeration toward –42 °C: brittle fracture risk, especially if re-pressured cold	Procedure	S4 L2 8 Medium

HAZOP recommendation

Check the minimum design metal temperature of V-201, V-202, C-201 and the LPG lines against auto-refrigeration: released or depressured LPG flashes toward its atmospheric boiling point (about –42 °C for propane). Depressure slowly, never re-pressure a cold vessel, add a low-temperature alarm on TI-201.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R07

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R07
Node	N1/N6/N9 · Feed surge drum V-201 and feed pumps P-201A/B / Reflux drum V-202, pumps P-202A/B and propane product LV-203 / Procedures: start-up, shutdown, emergency shutdown, maintenance	Action party	Process / Operations
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.8	Composition AS WELL AS : Free water in the feed	Carry-over from the upstream unit	Water settles in V-201: ice and hydrates at -1°C in drains and instrument taps; draining LPG with the water, a drain valve can freeze open	None shown (no water boot on the P&ID)	S4 L2 8 Medium
N6.8	Composition AS WELL AS : Water in V-202	- Water in the feed - condenser tube leak	Free water with the product; ice or hydrates on draining; drain valve can freeze open	None shown (no water boot)	S4 L2 8 Medium
N9.6	Water draining OTHER THAN : Drain valve freezes open while draining water	LPG follows the water through the drain and auto-refrigerates	Uncontrolled LPG release at the drain, vapour cloud, fire; historic LPG disasters started this way	Procedure	S5 L2 10 High

HAZOP recommendation

Water draining of the LPG drums: water boots with interface level on V-201 and V-202 (recommended addition), double-block drains with a spring-return valve, drains always attended; hydrate and ice awareness at -1°C .

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R08

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R08
Node	all · all	Action party	Instrument / Mechanical
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.9	Integrity OTHER THAN: Pump seal or flange leak	Seal failure (dry running, deadheading, gas-binding), gasket failure	LPG release at 2,363 kPa g: about 34 % flashes at once (model estimate), vapour cloud, fire or vapour-cloud explosion	Operator rounds	S5 L2 10 High
N3.9	Integrity OTHER THAN: Leak of hot bottoms	Flange, gasket or reboiler channel leak	C4 and heavier at 109 °C and 1,718 kPa g, above its atmospheric boiling point: almost all flashes, jet fire or vapour-cloud explosion	Operator rounds	S5 L2 10 High
N5.8	Temperature MORE: External fire at C-201 or V-202	Pool fire from an LPG leak at the pumps or the bottoms	Vapour generation beyond the process cases; wall overheating above the liquid, rupture and BLEVE	PSV-201 (fire case to be confirmed)	S5 L2 10 High
N6.2	Level LESS: V-202 empties	- LV-203 opens wide (LIC-203 transmitter reads high) - cooling water lost (nothing condenses)	Pumps lose suction and run dry: seal failure, LPG release; reflux lost (model, cooling water lost: LIC-203 LO at 04:20, level down to 0.3 %)	- LIC-203 LO alarm (25 %) - PDSL-230 starts the spare, which also runs dry - no low-low trip	S3 L3 9 Medium
N6.9	Integrity OTHER THAN: P-202A/B seal or flange leak	Seal failure after dry running or deadheading	LPG at 2,262 kPa g and 39 °C: about 75 % flashes at once (model estimate), vapour cloud, fire or explosion	Operator rounds	S5 L2 10 High
N9.7	ESD LATE: Emergency shutdown delayed	Operator unsure, many alarms at once	Leak or fire continues to be fed	ESD pushbutton (closes XV-201, XV-202, XV-209)	S4 L2 8 Medium

HAZOP recommendation

Gas detection around P-201A/B, P-202A/B, V-201, V-202, E-203 and the bottoms line, with remote-operated shut-off valves on the V-201 and V-202 outlets and on the column bottoms; double mechanical seals with seal-leak alarms on the LPG pumps.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R09

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R09
Node	N2 · Feed control FV-201 and preheater E-201 to tray 18	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.6	Pressure MORE : E-201 process side blocked in with steam on	Feed isolated for maintenance or by XV-201 while TV-202 stays open	Blocked-in LPG heated by 156 °C steam: pressure rises far above design, flange leak or rupture, LPG release	None on the P&ID (no thermal relief)	S4 L2 8 Medium

HAZOP recommendation

Thermal relief on the E-201 process side (recommended addition), or a locked procedure: steam off (XV-202, TV-202) before the feed is blocked in.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R10

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R10
Node	N2/N3 · Feed control FV-201 and preheater E-201 to tray 18 / C-201 stripping section (trays 1–17), sump and reboiler E-203	Action party	Process / Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.7	Integrity OTHER THAN: E-201 tube leak	Corrosion, vibration, thermal cycling	LPG at about 2,363 kPa g enters the steam side at 450 kPa g: hydrocarbon in the condensate return, fire or explosion in the condensate system or the boiler house	None (no detection on the condensate)	S4 L2 8 Medium
N3.10	Integrity OTHER THAN: E-203 tube leak	Corrosion, vibration	Hydrocarbon at 1,718 kPa g into the steam chest (254 kPa g): LPG in the condensate return	PI-205 rise (not alarmed)	S4 L2 8 Medium

HAZOP recommendation

Hydrocarbon detection on the E-201 and E-203 steam condensate: the process side is at higher pressure than the steam side, so a tube leak puts LPG into the condensate return. Route the condensate through a flash pot vented to a safe place.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R11

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R11
Node	N3/N4/N5 · C-201 stripping section (trays 1-17), sump and reboiler E-203 / C-201 rectifying section (trays 18-27) and reflux FV-204 / Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	Action party	Process safety / Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Heat MORE : Too much boil-up	- TV-208 fails open or TIC-208 forced to MAN high - TT-208 reads low	Heavies carried overhead (model, TIC-208 MAN 100 %: tray 8 to 110 °C, AI-219 25 % after 30 min), vapour load and pressure up (1.736 MPa g), flooding; relief if the condenser cannot follow	- TIC-208 HI alarm (82 °C) - AI-219 HI alarm (2.5 mol %) - TIC-227 HI alarm (50 °C) - PSV-201	S3 L3 9 Medium
N4.2	Flow MORE : Too much reflux	- FV-204 fails open (fail-open valve) - FIC-204 set point high - FT-204 reads low	Flooding: FIC-204 to 89 m³/h, PDI-212 to 112 kPa; TIC-208 adds steam to hold tray 8, pressure rises and PSV-201 lifts at 03:02 (up to about 2,210 Nm³/h to flare); AI-220 to 29 %	- FIC-204 HI alarm (65 m³/h) - PDI-212 HI alarm (30 kPa) - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201	S4 L3 12 High
N5.1	Pressure MORE : Overpressure: cooling water lost	- CW pump or cooling-tower failure (site-wide) - HV-216 closed	Nothing condenses: pressure climbs, PSV-201 lifts at 00:27 and the model peaks at 1.868 MPa g relieving up to about 19,900 Nm³/h (6,500 Nm³ in 30 min) with no trip; V-202 drains (LIC-203 LO at 04:20, down to 0.3 %) and the reflux pumps run dry	- FI-216 LO alarm (250 m³/h) (first, at 00:02) - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201 - no high-pressure trip	S5 L3 15 High
N5.2	Pressure MORE : Overpressure: vent blocked	- HV-210 closed by mistake - PV-210 fails closed or PIC-210 in MAN at low output - fuel-gas header blocked	Non-condensables accumulate and blanket E-202: pressure rises slowly (model, PV-210 closed: PIC-210 HI at 06:19, PI-211 HI at 07:37, PSV-201 lifts at 07:41, about 330 Nm³/h)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201	S4 L3 12 High
N5.3	Pressure MORE : Overpressure: condenser capacity reduced	- Warmer cooling water - condenser fouling	Higher condensing temperature: PV-210 vents more (and more propane with it), then the pressure rises to relief (model: CW 15 °C warmer lifts PSV-201 at 01:43, up to 6,100 Nm³/h; condenser UA –50 % lifts it at 01:52, vent up to 521 Nm³/h)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - TI-214, TI-215 - PSV-201	S4 L2 8 Medium
N5.6	Measurement OTHER THAN : PIC-210 transmitter wrong	Drift, impulse-line leak or blockage, calibration error	PV-210 acts on a wrong pressure: the true column pressure can move away from 1.70 MPa g in either direction without the control alarm seeing it	- PI-211 independent transmitter with its own HI alarm - PSV-201	S4 L2 8 Medium
N9.4	Heating EARLY : Heat raised too fast	Steam valves opened in large steps at start-up	Pressure surge, flooding, PSV-201 lift, tray damage	- PIC-210 alarms - procedure	S3 L3 9 Medium

HAZOP recommendation

High-pressure trip PSHH on two or three independent transmitters (recommended addition, 2003) that closes XV-209 and XV-202 (heat off) before PSV-201 lifts; LOPA to set its SIL. Today PSV-201 is the only layer against every overpressure scenario.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R12

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R12
Node	N3 · C-201 stripping section (trays 1-17), sump and reboiler E-203	Action party	Instrument / Operations
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.5	Temperature LESS : Feed colder, subcooled	- E-201 steam restricted - TV-202 sticks - condensate backing up in E-201 - low header pressure	More liquid to the stripping section and more reboiler duty: bottoms off spec (model, steam restricted: TIC-202 LO at 00:52; feed 63 °C; AI-220 up to 5.9 %)	- TIC-202 LO alarm (68 °C) - TIC-208 adds steam - AI-220 HI alarm (2 mol %)	S2 L3 6 Medium
N3.2	Heat LESS : Too little boil-up	- E-203 fouling - LP header pressure low - TV-208 sticks - condensate backing up in E-203 - HV-209 partly closed	C3 left in the bottoms: AI-220 off spec to the debutanizer (model, header falling to 180 kPa g: AI-220 23 %, first alarm PI-207 LO at 00:45); weeping at low vapour	- PI-207 LO alarm (350 kPa g) - TIC-208 LO alarm (72 °C) - AI-220 HI alarm (2 mol %) - PI-205 steam chest	S2 L4 8 Medium
N7.5	Composition OTHER THAN : Propane in the bottoms	- Too little boil-up (N3) - feed change	Debutanizer overhead overloaded with C3: its pressure rises; product losses	AI-220 HI alarm (2 mol %)	S2 L4 8 Medium
N8.3	Utility LESS : LP steam header pressure low	Boiler or header upset (toward 180 kPa g)	Less boil-up and preheat: bottoms off spec (model: PI-207 LO at 00:45, AI-220 HI at 03:26, up to 23 %)	- PI-207 LO alarm (350 kPa g) - TIC-208 LO alarm (72 °C) - AI-220 HI alarm (2 mol %)	S2 L4 8 Medium

HAZOP recommendation

Reboiler margin: alarm on TV-208 opening above 90 % and trend PI-205 (steam chest) so fouling, low header pressure and a stuck valve are seen before the bottoms go off spec.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R13

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R13
Node	N3/N7 · C-201 stripping section (trays 1-17), sump and reboiler E-203 / Bottoms to the debutanizer (LIC-202, LV-202)	Action party	Process / Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.4	Level LESS : Sump level low	• LV-202 sticks open • LIC-202 transmitter frozen or reading high • debutanizer pressure low	Reboiler circulation lost; then vapour at 1,718 kPa g blows through LV-202 into the debutanizer at 480 kPa g: overpressure downstream	• LIC-202 LO alarm (25 %) • LSLL-202 trip (sump < 5 %, 5 s) closes XV-209 and XV-202 (closes the steam only, not LV-202) • LG-202	S4 L2 8 Medium
N7.2	Flow MORE : Gas blow-by to the debutanizer	Sump emptied (LV-202 sticks open, LIC-202 transmitter frozen or reading high, bypass open)	Vapour at 1,718 kPa g flows into the debutanizer at 480 kPa g: overpressure of the debutanizer, relief there	• LIC-202 LO alarm (25 %) • LSLL-202 trip (sump < 5 %, 5 s) closes XV-209 and XV-202 (closes the steam, not LV-202)	S4 L2 8 Medium
N7.7	Measurement OTHER THAN : LIC-202 transmitter frozen	Transmitter or impulse-line fault	The reading looks steady while the true level drifts: the sump can empty (blow-by) or fill (flooding) without an alarm on LIC-202	• LSLL-202 (independent low-low switch) • LG-202	S4 L2 8 Medium

HAZOP recommendation

On low sump level close the bottoms line too: LSLL-202 today closes only the steam valves, so vapour at 1,718 kPa g can blow through LV-202 into the debutanizer at 480 kPa g. Add an XV on the bottoms line (recommended addition) and check the debutanizer relief for gas blow-by.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R14

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R14
Node	N3/N7 · C-201 stripping section (trays 1-17), sump and reboiler E-203 / Bottoms to the debutanizer (LIC-202, LV-202)	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.5	Level MORE : Sump level high	- LV-202 fails closed - LIC-202 transmitter frozen or reading low - debutanizer pressure high	Liquid backs up over the reboiler return and the bottom trays: flooding, PDI-212 up, off-spec products (model, LV-202 closed: LIC-202 HI alarm (75 %) at 04:21, PDI-212 HI at 19:57, no trip)	- LIC-202 HI alarm (75 %) - PDI-212 HI alarm (30 kPa) - no high-level trip	S3 L3 9 Medium
N7.1	Flow NO : No bottoms to the debutanizer	- LV-202 fails closed (fault or instrument air) - block valve closed	Debutanizer loses its feed; the C-201 sump fills (model: LIC-202 HI at 04:21, sump full, PDI-212 HI at 19:57, no trip)	- LIC-202 HI alarm (75 %) - PDI-212 HI alarm (30 kPa)	S3 L3 9 Medium
N7.3	Pressure MORE : Debutanizer pressure high	Debutanizer upset (toward 1.5 MPa g)	Less driving pressure: LV-202 opens further, the sump rises (model: 70 % with no alarm)	LIC-202 HI alarm (75 %)	S2 L3 6 Medium
N7.7	Measurement OTHER THAN : LIC-202 transmitter frozen	Transmitter or impulse-line fault	The reading looks steady while the true level drifts: the sump can empty (blow-by) or fill (flooding) without an alarm on LIC-202	- LSLL-202 (independent low-low switch) - LG-202	S4 L2 8 Medium

HAZOP recommendation

Independent high-high sump level switch LSHH-202 (recommended addition) with alarm, and a procedure to cut feed and heat when the sump cannot be emptied.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R15

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R15
Node	N3 · C-201 stripping section (trays 1-17), sump and reboiler E-203	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Heat MORE : Too much boil-up	- TV-208 fails open or TIC-208 forced to MAN high - TT-208 reads low	Heavies carried overhead (model, TIC-208 MAN 100 %: tray 8 to 110 °C, AI-219 25 % after 30 min), vapour load and pressure up (1.736 MPa g), flooding; relief if the condenser cannot follow	- TIC-208 HI alarm (82 °C) - AI-219 HI alarm (2.5 mol %) - TIC-227 HI alarm (50 °C) - PSV-201	S3 L3 9 Medium
N3.8	Measurement OTHER THAN : TT-208 (tray 8) reads wrong	Drift, thermowell fouling, wiring	TIC-208 moves the steam the wrong way: bottoms off spec (reads high) or heavies overhead and pressure up (reads low)	- TI-242, TI-243 nearby - AI-220 HI alarm (2 mol %) - AI-219 HI alarm (2.5 mol %)	S2 L3 6 Medium

HAZOP recommendation

Deviation alarm between TIC-208 (tray 8) and the neighbouring TI-242 (tray 4) and TI-243 (tray 12) so a drifting control transmitter is seen.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R16

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R16
Node	N1/N4/N6 · Feed surge drum V-201 and feed pumps P-201A/B / C-201 rectifying section (trays 18-27) and reflux FV-204 / Reflux drum V-202, pumps P-202A/B and propane product LV-203	Action party	Operations / Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.3	Flow NO : No feed from the pumps	• P-201A trips • P-201B turbine will not start (governor, steam) • suction valve closed	No feed to C-201: the column runs down to total reflux, pressure falls (1.32 MPa g in the model) and product flows stop; V-201 fills (row above)	• FSLL-201 (< 15 m³/h, 5 s) auto-starts P-201B • TIC-202 HI alarm (85 °C) (the first alarm in the model, at 00:02) • P-201B spare	S3 L3 9 Medium
N4.1	Flow NO : No reflux	• P-202A trips and P-202B does not start • FV-204 closed • V-202 empty (see N6)	Top tray heats, heavies go overhead (model: FIC-204 LO alarm (35 m ³ /h) at 00:01, TIC-227 HI at 01:01, AI-219 up to 9.6 %); the condensate stays in V-202, which overfills and trips LSHH-203 at 05:07	• PDSL-230 (< 150 kPa, 3 s) auto-starts P-202B • FIC-204 LO alarm (35 m³/h) • TIC-227 HI alarm (50 °C) • LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202	S3 L3 9 Medium
N6.3	Pressure LESS : Reflux pump ΔP low	• P-202A trips • pump gas-bound	Reflux and product interrupted; with P-202B starting automatically the model dips the reflux to 10 m ³ /h and recovers (alarms: FIC-204 LO 00:01)	• PDSL-230 (< 150 kPa, 3 s) auto-starts P-202B • FIC-204 LO alarm (35 m³/h)	S2 L3 6 Medium

HAZOP recommendation

Test the auto-start of P-201B (FSLL-201) and P-202B (PDSL-230) at a fixed interval and after maintenance; record the tests.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R17

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R17
Node	N4 · C-201 rectifying section (trays 18–27) and reflux FV-204	Action party	Process / Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.2	Flow MORE : Too much reflux	- FV-204 fails open (fail-open valve) - FIC-204 set point high - FT-204 reads low	Flooding: FIC-204 to 89 m ³ /h, PDI-212 to 112 kPa; TIC-208 adds steam to hold tray 8, pressure rises and PSV-201 lifts at 03:02 (up to about 2,210 Nm ³ /h to flare); AI-220 to 29 %	- FIC-204 HI alarm (65 m³/h) - PDI-212 HI alarm (30 kPa) - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201	S4 L3 12 High
N4.4	Pressure MORE : Rectifying ΔP high (flooding)	- Feed too hot or too much feed - too much boil-up - reflux too high	Entrainment, efficiency lost, both products off spec; ΔP swings	PDI-212 HI alarm (30 kPa)	S2 L3 6 Medium

HAZOP recommendation

Reflux too high: FIC-204 HI alarm response note and a high limit on the FIC-204 set point; review the fail-open choice of FV-204 (the model floods the column and lifts PSV-201 when it fails open).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R18

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R18
Node	N4/N6/N7 · C-201 rectifying section (trays 18-27) and reflux FV-204 / Reflux drum V-202, pumps P-202A/B and propane product LV-203 / Bottoms to the debutanizer (LIC-202, LV-202)	Action party	Laboratory / Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.1	Flow MORE : More feed than the column is designed for	- FIC-201 set point raised - FT-201 reads low - upstream pushes more	Column load above design: rectifying flooding, off-spec products, higher condenser duty; drum inventory changes (no level controller in rev A)	- PDI-212 HI alarm (30 kPa) - TIC-208 LO alarm (72 °C) - AI-220 HI alarm (2 mol %) (no feed-flow alarm in rev A)	S2 L3 6 Medium
N2.8	Composition OTHER THAN : Feed composition changes	Upstream unit changes feedstock or operation	Lighter feed: more propane, AI-219 up to 1.9 % in the model; heavier feed: AI-220 up to 3.1 % (AI-220 HI at 14:22)	- AI-219 HI alarm (2.5 mol %) - AI-220 HI alarm (2 mol %) - TIC-208 moves the steam	S2 L4 8 Medium
N3.6	Pressure MORE : Stripping section ΔP high (foaming or flooding)	- Foaming (feed contaminants, additives) - vapour load above flood	Entrainment, efficiency lost, bottoms off spec (model, foaming: PDI-212 HI at 00:42, up to 90 kPa; AI-220 7.6 %)	- PDI-212 HI alarm (30 kPa) - AI-220 HI alarm (2 mol %)	S2 L3 6 Medium
N3.7	Efficiency LESS : Stages lost (tray damage)	Pressure surge, water slug or corrosion lifts or damages stripping trays	Same boil-up strips less: bottoms off spec without a ΔP rise	- AI-220 HI alarm (2 mol %) - tray temperatures	S2 L2 4 Low
N4.3	Flow LESS : Too little reflux	- FIC-204 set point low - FV-204 sticks - pump wear	Heavies overhead: propane product off spec	- AI-219 HI alarm (2.5 mol %) - TIC-227 HI alarm (50 °C)	S2 L3 6 Medium
N4.6	Composition OTHER THAN : Propane product off spec	- Any of the rows above - lighter or heavier feed - analyzer fault	Product downgraded (330 instead of 420 \$/m ³ in the training prices); off-spec propane into the sales tank	AI-219 HI alarm (2.5 mol %)	S2 L4 8 Medium
N6.6	Composition OTHER THAN : Off-spec propane to storage	- Heavies overhead (N4) - analyzer fault	Sales tank contaminated, product downgraded	AI-219 HI alarm (2.5 mol %)	S2 L3 6 Medium
N7.5	Composition OTHER THAN : Propane in the bottoms	- Too little boil-up (N3) - feed change	Debutanizer overhead overloaded with C3: its pressure rises; product losses	AI-220 HI alarm (2 mol %)	S2 L4 8 Medium

HAZOP recommendation

Validate AI-219 and AI-220 against a laboratory sample every shift; alarm on analyzer fault.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			

Sign-off	Name	Signature	Date
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R19

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R19
Node	N4 · C-201 rectifying section (trays 18-27) and reflux FV-204	Action party	Process control
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.7	Control OTHER THAN : Two composition loops fight	TIC-227 cascade and TIC-208 both on automatic (interacting loops)	Reflux and boil-up cycle against each other; both products swing	TIC-227 normally in MAN	S2 L2 4 Low

HAZOP recommendation

Composition control: run the TIC-227 → FIC-204 cascade and TIC-208 together only after a decoupling review; operating note on which loop to put in MAN first.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R20

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R20
Node	N5/N8 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201 / Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	Action party	Process safety
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.1	Pressure MORE : Overpressure: cooling water lost	- CW pump or cooling-tower failure (site-wide) - HV-216 closed	Nothing condenses: pressure climbs, PSV-201 lifts at 00:27 and the model peaks at 1.868 MPa g relieving up to about 19,900 Nm ³ /h (6,500 Nm ³ in 30 min) with no trip; V-202 drains (LIC-203 LO at 04:20, down to 0.3 %) and the reflux pumps run dry	- FI-216 LO alarm (250 m³/h) (first, at 00:02) - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201 - no high-pressure trip	S5 L3 15 High
N5.3	Pressure MORE : Overpressure: condenser capacity reduced	- Warmer cooling water - condenser fouling	Higher condensing temperature: PV-210 vents more (and more propane with it), then the pressure rises to relief (model: CW 15 °C warmer lifts PSV-201 at 01:43, up to 6,100 Nm ³ /h; condenser UA –50 % lifts it at 01:52, vent up to 521 Nm ³ /h)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - TI-214, TI-215 - PSV-201	S4 L2 8 Medium
N6.7	Temperature MORE : Reflux drum hot	- Cooling water lost or warm - condenser fouled	Propane vapour pressure rises with temperature (1,908 kPa g at 50 °C in the model, above the PSV-201 set pressure): the drum is held up at relief pressure through the column; reflux pumps lose NPSH	- TI-213 - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201 (if V-202 cannot be isolated from C-201)	S3 L3 9 Medium
N8.2	Utility NO : Cooling water lost (site-wide)	Cooling-tower pumps or power lost	See N5: relief of C-201 to flare, and of every other column on the site at the same time	- FI-216 LO alarm (250 m³/h) - PSV-201	S5 L2 10 High
N8.7	Utility MORE : Flare header back-pressure high	Several units relieving together (CW loss)	PSV-201 relieves less than its rating: higher column peak pressure	Flare design (outside scope)	S4 L2 8 Medium

HAZOP recommendation

Flare study: loss of cooling water is common to every column on the site, so check the flare header and flare for the simultaneous relief (C-201 alone reaches about 19,900 Nm³/h in the model); back-pressure on PSV-201.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			

Sign-off	Name	Signature	Date
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R21

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R21
Node	N5/N6 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201 / Reflux drum V-202, pumps P-202A/B and propane product LV-203	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.1	Pressure MORE : Overpressure: cooling water lost	- CW pump or cooling-tower failure (site-wide) - HV-216 closed	Nothing condenses: pressure climbs, PSV-201 lifts at 00:27 and the model peaks at 1.868 MPa g relieving up to about 19,900 Nm ³ /h (6,500 Nm ³ in 30 min) with no trip; V-202 drains (LIC-203 LO at 04:20, down to 0.3 %) and the reflux pumps run dry	- FI-216 LO alarm (250 m³/h) (first, at 00:02) - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201 - no high-pressure trip	S5 L3 15 High
N6.2	Level LESS : V-202 empties	- LV-203 opens wide (LIC-203 transmitter reads high) - cooling water lost (nothing condenses)	Pumps lose suction and run dry: seal failure, LPG release; reflux lost (model, cooling water lost: LIC-203 LO at 04:20, level down to 0.3 %)	- LIC-203 LO alarm (25 %) - PDSL-230 starts the spare, which also runs dry - no low-low trip	S3 L3 9 Medium

HAZOP recommendation

Low-low level on V-202 (LSLL-203, recommended addition) that stops P-202A/B before they run dry.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R22

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R22
Node	N5 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	Action party	Operations / Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.2	Pressure MORE : Overpressure: vent blocked	- HV-210 closed by mistake - PV-210 fails closed or PIC-210 in MAN at low output - fuel-gas header blocked	Non-condensables accumulate and blanket E-202: pressure rises slowly (model, PV-210 closed: PIC-210 HI at 06:19, PI-211 HI at 07:37, PSV-201 lifts at 07:41, about 330 Nm ³ /h)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201	S4 L3 12 High

HAZOP recommendation

Car-seal HV-210 open (CSO register) and add a PIC-210 output-high alarm: a vent that cannot open leaves the column to PSV-201.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R23

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R23
Node	N5/N8 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201 / Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.4	Composition AS WELL AS : More non-condensables	Methane or hydrogen from the upstream unit	Condenser area blanketed; PV-210 vents more gas and propane (model, five-fold: vent up to 361 Nm ³ /h, PIC-210 HI at 03:29, PSV-201 briefly lifts at 04:17)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter	S3 L3 9 Medium
N5.5	Flow MORE : Too much vent to fuel gas (column depressured)	PV-210 fails open (instrument air) or PIC-210 forced high	Propane vapour lost to fuel gas (model, PV-210 wide open: vent up to 483 Nm ³ /h, pressure down to 1.54 MPa g, PIC-210 LO at 05:20); fuel-gas header overpressure; column upset	- PIC-210 LO alarm (1.6 MPa g) - fuel-gas system relief	S3 L3 9 Medium
N6.1	Level MORE : V-202 overfills	- LV-203 fails closed or storage blocked in - LIC-203 in MAN at low output - reflux lost	Liquid LPG rises into the vent to the fuel-gas header (burner flooding, fire at the users). Model, LV-203 closed: LIC-203 HI alarm (75 %) at 09:00, LSHH-203 trips at 13:43 (feed and heat off)	- LIC-203 HI alarm (75 %) - LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202	S4 L2 8 Medium
N8.6	Utility MORE : Fuel-gas header pressure high	Fuel-gas users shut down	Less vent through PV-210: pressure rises as in a blocked vent	- PIC-210 HI alarm (1.76 MPa g) - PSV-201	S3 L2 6 Medium

HAZOP recommendation

Fuel-gas interface: knock-out drum on the PV-210 vent before the fuel-gas header, high-flow alarm on FI-217, and tell the fuel-gas system owner that PV-210 fails open.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R24

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R24
Node	N5 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.6	Measurement OTHER THAN: PIC-210 transmitter wrong	Drift, impulse-line leak or blockage, calibration error	PV-210 acts on a wrong pressure: the true column pressure can move away from 1.70 MPa g in either direction without the control alarm seeing it	- PI-211 independent transmitter with its own HI alarm - PSV-201	S4 L2 8 Medium

HAZOP recommendation

Deviation alarm between PIC-210 and PI-211: two independent transmitters must agree; operating note on which to believe.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R25

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R25
Node	N5 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	Action party	Mechanical / Operations
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.7	Relief OTHER THAN : PSV-201 unavailable or passing	- Inlet or outlet isolation closed - set too high - seat damage after lifting	No protection against the overpressure rows above (vessel rupture), or continuous loss of propane to flare	Relief valve test programme	S5 L1 5 Medium

HAZOP recommendation

PSV-201 inlet and outlet isolation valves car-sealed open; test and inspection interval; flare line free-draining.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R26

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R26
Node	N5 · Overhead: condenser E-202, vent PV-210 to fuel gas, pressure and PSV-201	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.9	Integrity OTHER THAN: E-202 tube leak	Corrosion, vibration	LPG at 1.70 MPa g into the cooling water: flammable gas released at the cooling tower	None	S4 L2 8 Medium
N6.8	Composition AS WELL AS: Water in V-202	- Water in the feed - condenser tube leak	Free water with the product; ice or hydrates on draining; drain valve can freeze open	None shown (no water boot)	S4 L2 8 Medium

HAZOP recommendation

Hydrocarbon detection on the E-202 cooling-water return: LPG leaking into the cooling water is released at the cooling tower.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R27

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R27
Node	N6 · Reflux drum V-202, pumps P-202A/B and propane product LV-203	Action party	Instrument / Process
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.1	Level MORE : V-202 overfills	- LV-203 fails closed or storage blocked in - LIC-203 in MAN at low output - reflux lost	Liquid LPG rises into the vent to the fuel-gas header (burner flooding, fire at the users). Model, LV-203 closed: LIC-203 HI alarm (75 %) at 09:00, LSHH-203 trips at 13:43 (feed and heat off)	- LIC-203 HI alarm (75 %) - LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202	S4 L2 8 Medium

HAZOP recommendation

Let LSHH-203 also close PV-210 (or confirm the knock-out drum of R23) so liquid LPG cannot reach the fuel-gas header when V-202 overfills.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R28

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R28
Node	N6 · Reflux drum V-202, pumps P-202A/B and propane product LV-203	Action party	Process / Operations
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.6	Composition OTHER THAN : Propane product off spec	- Any of the rows above - lighter or heavier feed - analyzer fault	Product downgraded (330 instead of 420 \$/m ³ in the training prices); off-spec propane into the sales tank	AI-219 HI alarm (2.5 mol %)	S2 L4 8 Medium
N6.6	Composition OTHER THAN : Off-spec propane to storage	- Heavies overhead (N4) - analyzer fault	Sales tank contaminated, product downgraded	AI-219 HI alarm (2.5 mol %)	S2 L3 6 Medium

HAZOP recommendation

Off-spec propane: divert the product to an off-spec tank on AI-219 HI instead of the sales tank.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R29

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R29
Node	N8 · Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	Action party	Instrument / Process
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.5	Flow MORE : Too much vent to fuel gas (column depressured)	PV-210 fails open (instrument air) or PIC-210 forced high	Propane vapour lost to fuel gas (model, PV-210 wide open: vent up to 483 Nm ³ /h, pressure down to 1.54 MPa g, PIC-210 LO at 05:20); fuel-gas header overpressure; column upset	- PIC-210 LO alarm (1.6 MPa g) - fuel-gas system relief	S3 L3 9 Medium
N8.1	Utility NO : Instrument air lost	Compressor or dryer failure, header break	Every control valve goes to its fail position (as marked on the P&ID). Model: the column depressures to 0.83 MPa g through PV-210 into the fuel gas, the sump fills, V-202 empties to 0.3 %, V-201 goes liquid-full (95 % after about 12 min), no trip	- Fail positions - alarms on many loops at once	S3 L3 9 Medium

HAZOP recommendation

Instrument-air failure: review the fail positions (PV-210 fails open to fuel gas, LV-202 and LV-203 fail closed) and consider an ESD on low air-header pressure; confirm the air receiver hold-up time.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R30

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R30
Node	N8 · Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	Action party	Electrical / Operations
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.5	Utility NO : Power lost	Grid or substation failure	P-201A and P-202A/B stop (P-201B is a steam turbine): reflux lost, V-202 overfills and LSHH-203 trips (as in N4 no reflux)	- LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202 - P-201B on steam	S3 L3 9 Medium

HAZOP recommendation

Power failure: supply P-202B from a different bus (or a steam driver like P-201B) and write the power-failure response procedure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R31

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R31
Node	N8 · Utilities: LP steam, cooling water, instrument air, power, fuel gas and flare	Action party	Mechanical
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.4	Utility MORE : LP steam header pressure high	Let-down valve fails open upstream	E-201 and E-203 steam sides above their design pressure; more heat available if TV-208/TV-202 open	Header relief (outside scope)	S3 L2 6 Medium

HAZOP recommendation

Confirm the steam-side design pressure of E-201 and E-203 against the highest LP header pressure, and their relief.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R32

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R32
Node	N9 · Procedures: start-up, shutdown, emergency shutdown, maintenance	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.3	Purging PART OF : Air left in the system at start-up	Nitrogen purge incomplete or skipped	LPG and air form a flammable mixture inside C-201, V-202 or the flare line: internal explosion	Procedure (no oxygen check point on the P&ID)	S5 L2 10 High

HAZOP recommendation

Start-up: purge C-201, V-202, V-201 and the lines with nitrogen to an oxygen content below the limit before hydrocarbon is admitted; record the gas tests; hold point in the procedure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R33

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R33
Node	N9 · Procedures: start-up, shutdown, emergency shutdown, maintenance	Action party	Operations / Safety
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.8	Maintenance OTHER THAN : Vessel entry at turnaround	- Nitrogen or hydrocarbon not cleared - isolation incomplete	Asphyxiation or fire inside the vessel	- Permit to work - gas test	S4 L2 8 Medium

HAZOP recommendation

Turnaround entry: positive isolation (blinds), nitrogen disconnected, oxygen and hydrocarbon tests, standby person and rescue plan for every vessel entry.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R34

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R34
Node	N9 · Procedures: start-up, shutdown, emergency shutdown, maintenance	Action party	Operations / Instrument
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.9	Safety system OTHER THAN : Trip bypassed	LSSL-202 or LSHH-203 bypassed for testing or nuisance trips	No protection against reboiler dry-out or liquid to fuel gas	Permit to work	S4 L2 8 Medium

HAZOP recommendation

Trip-bypass management for LSSL-202, LSHH-203 and the auto-starts: permit, time limit, bypass shown on the DCS, proof tests at the interval set by LOPA.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R35

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R35
Node	N9 · Procedures: start-up, shutdown, emergency shutdown, maintenance	Action party	Mechanical / Operations
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.10	Steam-out AFTER : Vessel closed after steaming out	Steam condenses in a closed column or drum as it cools	Vacuum: shell or tray damage if not vacuum-rated	Procedure	S4 L2 8 Medium

HAZOP recommendation

Steam-out: confirm the vacuum rating of C-201 and V-202, or keep a vent open while the steamed vessel cools.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R36

Project	Depropanizer Column Ver 1 · C-201 Operator Training Simulator	Document	DC3-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R36
Node	all · all	Action party	Process safety
Drawings	P&ID-001 rev C (screen drawing); DC3-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.6	Pressure MORE : V-201 blocked in and heated	Sun or external fire on an isolated drum	Vapour pressure rises with temperature (1,037 kPa g at 40 °C in the model); in a fire the wall above the liquid overheats: vessel rupture and BLEVE with fireball	None on the P&ID (no relief valve on V-201)	S5 L2 10 High
N1.9	Integrity OTHER THAN : Pump seal or flange leak	Seal failure (dry running, deadheading, gas-binding), gasket failure	LPG release at 2,363 kPa g: about 34 % flashes at once (model estimate), vapour cloud, fire or vapour-cloud explosion	Operator rounds	S5 L2 10 High
N3.9	Integrity OTHER THAN : Leak of hot bottoms	Flange, gasket or reboiler channel leak	C4 and heavier at 109 °C and 1,718 kPa g, above its atmospheric boiling point: almost all flashes, jet fire or vapour-cloud explosion	Operator rounds	S5 L2 10 High
N4.1	Flow NO : No reflux	- P-202A trips and P-202B does not start - FV-204 closed - V-202 empty (see N6)	Top tray heats, heavies go overhead (model: FIC-204 LO alarm (35 m³/h) at 00:01, TIC-227 HI at 01:01, AI-219 up to 9.6 %); the condensate stays in V-202, which overfills and trips LSHH-203 at 05:07	- PDSL-230 (< 150 kPa, 3 s) auto-starts P-202B - FIC-204 LO alarm (35 m³/h) - TIC-227 HI alarm (50 °C) - LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202	S3 L3 9 Medium
N4.2	Flow MORE : Too much reflux	- FV-204 fails open (fail-open valve) - FIC-204 set point high - FT-204 reads low	Flooding: FIC-204 to 89 m³/h, PDI-212 to 112 kPa; TIC-208 adds steam to hold tray 8, pressure rises and PSV-201 lifts at 03:02 (up to about 2,210 Nm³/h to flare); AI-220 to 29 %	- FIC-204 HI alarm (65 m³/h) - PDI-212 HI alarm (30 kPa) - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201	S4 L3 12 High
N5.1	Pressure MORE : Overpressure: cooling water lost	- CW pump or cooling-tower failure (site-wide) - HV-216 closed	Nothing condenses: pressure climbs, PSV-201 lifts at 00:27 and the model peaks at 1.868 MPa g relieving up to about 19,900 Nm³/h (6,500 Nm³ in 30 min) with no trip; V-202 drains (LIC-203 LO at 04:20, down to 0.3 %) and the reflux pumps run dry	- FI-216 LO alarm (250 m³/h) (first, at 00:02) - PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201 - no high-pressure trip	S5 L3 15 High
N5.2	Pressure MORE : Overpressure: vent blocked	- HV-210 closed by mistake - PV-210 fails closed or PIC-210 in MAN at low output - fuel-gas header blocked	Non-condensables accumulate and blanket E-202: pressure rises slowly (model, PV-210 closed: PIC-210 HI at 06:19, PI-211 HI at 07:37, PSV-201 lifts at 07:41, about 330 Nm³/h)	- PIC-210 HI alarm (1.76 MPa g) - PI-211 HI alarm (1.78 MPa g) on the independent transmitter - PSV-201	S4 L3 12 High
N5.7	Relief OTHER THAN : PSV-201 unavailable or passing	- Inlet or outlet isolation closed - set too high - seat damage after lifting	No protection against the overpressure rows above (vessel rupture), or continuous loss of propane to flare	Relief valve test programme	S5 L1 5 Medium
N6.1	Level MORE : V-202 overfills	- LV-203 fails closed or storage blocked in - LIC-203 in MAN at low output - reflux lost	Liquid LPG rises into the vent to the fuel-gas header (burner flooding, fire at the users). Model, LV-203 closed: LIC-203 HI alarm (75 %) at 09:00, LSHH-203 trips at 13:43 (feed and heat off)	- LIC-203 HI alarm (75 %) - LSHH-203 trip (V-202 > 92 %, 5 s) closes XV-201, XV-209 and XV-202	S4 L2 8 Medium
N6.9	Integrity OTHER THAN : P-202A/B seal or flange leak	Seal failure after dry running or deadheading	LPG at 2,262 kPa g and 39 °C: about 75 % flashes at once (model estimate), vapour cloud, fire or explosion	Operator rounds	S5 L2 10 High
N7.2	Flow MORE : Gas blow-by to the debutanizer	Sump emptied (LV-202 sticks open, LIC-202 transmitter frozen or reading high, bypass open)	Vapour at 1,718 kPa g flows into the debutanizer at 480 kPa g: overpressure of the debutanizer, relief there	- LIC-202 LO alarm (25 %) - LSLL-202 trip (sump < 5 %, 5 s) closes XV-209 and XV-202 (closes the steam, not LV-202)	S4 L2 8 Medium

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.1	Utility NO : Instrument air lost	Compressor or dryer failure, header break	Every control valve goes to its fail position (as marked on the P&ID). Model: the column depressures to 0.83 MPa g through PV-210 into the fuel gas, the sump fills, V-202 empties to 0.3 %, V-201 goes liquid-full (95 % after about 12 min), no trip	• Fail positions • alarms on many loops at once	S3 L3 9 Medium
N9.3	Purging PART OF : Air left in the system at start-up	Nitrogen purge incomplete or skipped	LPG and air form a flammable mixture inside C-201, V-202 or the flare line: internal explosion	Procedure (no oxygen check point on the P&ID)	S5 L2 10 High
N9.6	Water draining OTHER THAN : Drain valve freezes open while draining water	LPG follows the water through the drain and auto-refrigerates	Uncontrolled LPG release at the drain, vapour cloud, fire; historic LPG disasters started this way	Procedure	S5 L2 10 High

HAZOP recommendation

LOPA for the High rows that rely on PSV-201, LSHH-203, LSL-202 and on the operator (overpressure from cooling-water loss, a blocked vent or too much reflux; V-201 overfill and fire; LPG leaks).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			